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When Mathematics Meets the Physical Trades

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• **OpenSCAD & KiCad Hardware Files:** Printable calculator enclosures, tactile keycap matrices, 42mm Gridfinity math blocks, and PCB schematics.

THE STACKCALC LIBRARY • VOLUME III

THE STACK IN THE FIELD

When Mathematics Meets the Physical Trades

Applied Field Mathematics, Domain Softkeys, and Four-Register Stack Traces Across Nine Professional Instruments (**SC-35, SC-14, SC-16, SC-24, SC-50, SC-60, SC-78, SC-80 & SC-95**)

StackCalc
stackcalc.io
First Edition

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For the surveyors, electricians, systems programmers, focus pullers, RF engineers, chemists, bakers, brewmasters, clinical nurses, and analysts
who put mathematics to work in the physical world.

"Theory without practice is empty;
practice without theory is blind."

— **IMMANUEL KANT, *Critique of Pure Reason* (1781)**

ARCHITECTURAL FRONTISPIECE: THE NINE DOMAIN INSTRUMENTS

Invariant Four-Register RPN Core, Row 0 Trade Softkeys & Domain Input Modes

1. One Invariant Stack Across Nine Physical Trades: Every calculator in the StackCalc Domain Series (**SC-35, SC-14, SC-16, SC-24, SC-50, SC-60, SC-78, SC-80,** and **SC-95**) executes inside the same four-level operational stack (**Registers T, Z, Y, X**). Learning RPN stack mechanics once grants immediate fluency across all nine instruments.

2. Row 0 Domain Softkey Solvers: Rather than burying trade formulas inside nested menus, each instrument maps its top physical row of six keys (**Row 0**) to the fundamental nouns and verbs of that discipline—such as **FWD**, **INV**, and **BRG** on the SC-35 Surveyor; **RUN**, **RISE**, **DIAG**, and **BEND** on the SC-14 Tradesman; **HEX**, **BIN**, **WSZ**, and **FLT** on the SC-16 Programmer; and **BSA**, **IBW**, **CrCl**, and **MCG** on the SC-80 Clinician.

3. Native Trade Input Modes: Numbers are entered and displayed in the exact notation of the field: packed sexagesimal degrees-minutes-seconds (**D.MMSS**) in surveying, feet-inches-sixteenths (**FT-IN**) in carpentry and conduit bending, 8/16/32/64-bit two's complement hexadecimal and binary in embedded systems, and frame-accurate **HH:MM:SS:FF** SMPTE timecode on cinema sets.

4. Worked Stack Traces & Verification: Every chapter pairs historical context and first-principles mathematical derivations with step-by-step four-register stack trace tables and self-check practice problems.

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StackCalc Domain Calculator Series

When Mathematics Meets the Physical Trades

Who this series is for: You already know how algebra, trigonometry, vectors, and logarithms work on paper. But how does an electrician turn right triangles into bent steel pipes thirty feet in the air? How does an artisan baker use percentages where the total is 197%? Why does an ICU nurse use square roots of height and weight to protect an elderly patient's kidneys? And why does a radio engineer care about waves reflecting backward down a coaxial cable?

The Philosophy: Mathematics Grounded in Physical Reality

Pure mathematics is clean, timeless, and abstract. But when math steps out of the lecture hall and onto a job site, it hits the friction of the real world:

- Heat dissipates in copper wires.
- Measurement errors compound across property lines.
- Microprocessors overflow when numbers exceed silicon registers.
- Yeast fermentation speeds up when a mixer blade gets warm.
- Sound waves collide in echoes and phase cancellations.

For decades, professionals in these fields used specialized pocket slide rules, tables, or legendary instruments like the **HP-35**, **HP-16C**, and **HP-12C**.

The **StackCalc Domain Series** expands that heritage into nine specialized physical and software instruments. Each calculator takes the timeless power of the 4-level Reverse Polish Notation (RPN) stack and customizes its keys, constants, and display for the natural rhythm of a specific craft.

Every instrument in the series shares five core engineering principles:

1. **Pure Scalar Stack (X, Y:, Z:, T:):** Calculations are never blocked by clunky unit-mismatch errors. The numbers are clean, double-precision scalars.
2. **Top-Zero Suppression:** Empty stack registers above your active numbers stay silent, keeping the screen uncluttered.
3. **Dedicated Table Inspector (TBL via [↷] [+/−]):** Lookup tables (AWG wire ampacities, periodic table molar masses, hop yields, and pipe dimensions) are accessible instantly without modifying your working stack.
4. **Physical Row 0 Softkeys:** Six dedicated keys align with LCD screen slots for 1-tap parameter workspaces.
5. **Non-Destructive Inspection (SHOW):** Transiently displays deep analytical breakdowns (like quadrant bearings, IEEE-754 float bitfields, or kitchen utensil equivalents) and returns to the stack untouched.

Discipline Chapters & Overviews

Chapter 1: SC-35 Geomatics, Land Surveying & Flight Navigation

Mapping a World That Refuses to Stand Still

- **The Core Puzzle:** Why do surveyors swap sine and cosine compared to standard math, and why do historic property deeds use quadrant bearings (S 45° W) instead of 0–360° angles?
- **Key Concepts:** Sexagesimal *DDD.MMSS* angles, Forward and Inverse Coordinate Geometry (COGO), traverse loop closure with Gauss's Shoelace polygon area, and solving aerodynamic crosswind drift.

Chapter 2: SC-14 Industrial Trades, Electrical & Mechanical Handbook

When Ohm's Law Meets Heavy Copper

- **The Core Puzzle:** Why does a vacuum cleaner dim your living room lamp, and how do electricians use cosecant multipliers to bend rigid steel conduit around ceiling beams?
- **Key Concepts:** Conductor resistance and Circular Mils (CM), single-phase and 3-phase voltage drop ($VD = 2KIL/CM$), the geometric $\sqrt{3} \approx 1.732$ factor of 3-phase AC power, and CNC surface feet per minute (SFM).

Chapter 3: SC-16 Systems Programming, Embedded & Cyber Operations

When Numbers Hit the Silicon Ceiling

- **The Core Puzzle:** Why do digital odometers and computer registers roll over to zero, and how did a 16-bit integer overflow destroy the Ariane 5 rocket?
- **Key Concepts:** Two's complement signed integer mechanics, speaking in hexadecimal (base-16 nibbles), memory alignment padding gaps, and dissecting single-precision IEEE-754 floating-point bits.

Chapter 4: SC-24 Cinema Optics, Broadcast Timecode & Sound Handbook

When Light and Sound Become Numbers

- **The Core Puzzle:** Why does North American television run at 29.97 frames per second instead of 30.00, and why are two people shouting together only 3 dB louder than one?
- **Key Concepts:** SMPTE drop-frame drift compensation, circle of confusion and hyperfocal depth-of-field, Exposure Value (EV) photometrics, and logarithmic acoustic decibel summing ($dB\oplus$).

Chapter 5: SC-50 RF Spectrum, Transmission Lines & Microwave Handbook

When Wires Stop Acting Like Wires

- **The Core Puzzle:** Why do high-frequency radio waves bounce backward down an antenna cable like an echo in a canyon, and how does a mismatched antenna melt an expensive transmitter transistor?
- **Key Concepts:** Characteristic impedance (50 Ω vs. 75 Ω), reflection coefficient (Γ), Voltage Standing Wave Ratio ($VSWR$), folding the infinite impedance plane into a Smith Chart circle, and Friis cascaded receiver noise.

Chapter 6: SC-60 Wet-Lab Chemistry, Stoichiometry & Spectroscopy

Counting What You Cannot See

- **The Core Puzzle:** How do analytical chemists translate an exact count of trillions of invisible molecules into a pile of white powder on a lab balance?
- **Key Concepts:** IUPAC molar mass summation ($m = C \cdot V \cdot MW$), serial dilution ($C_1 V_1 = C_2 V_2$), logarithmic pH scales, Henderson-Hasselbalch biochemical buffer design, and the Beer-Lambert spectrophotometry law.

Chapter 7: SC-78 Commercial Bakery, Fermentation Science & Brewing

When the Kitchen Becomes a Laboratory

- **The Core Puzzle:** Why is measuring flour by volume a trap, and why do commercial bakeries anchor flour at 100% so that recipes can total 197%?
- **Key Concepts:** Baker's Percentages ($BK\%$), Desired Dough Water Temperature (T_W) with spiral-mixer friction compensation, equilibrium meat curing salinity, and brewing specific gravity (SG, °P, ABV).

Chapter 8: SC-80 Clinical Nursing, Hospital Pharmacy & Infusions

When a Decimal Point Means Everything

- **The Core Puzzle:** Why is dosing a drug purely by body weight dangerous, and why do cancer chemotherapy and critical care medications scale with surface area (m^2)?
- **Key Concepts:** The persistent Patient Profile (**@PAT**), Mosteller Body Surface Area, Cockcroft-Gault Creatinine Clearance ($CrCl$) for aging kidneys, and microgram weight-based infusion pump titrations (µg/kg/min).

Chapter 9: SC-95 Quantitative Finance, Actuarial Science & Six Sigma

The Cost of Tomorrow: Time, Uncertainty, and Capital

- **The Core Puzzle:** Why is $10,000 today worth more than $10,000 in ten years, and how do industrial engineers know if a factory process is producing defect-free parts?
- **Key Concepts:** The William Kahan 5-key TVM system (PV, i, N, PMT, FV), discounted cash flows (NPV and IRR), bond pricing, and Six Sigma process capability indices (C_p, C_{pk}).

SC-35 Geomatics, Land Surveying & Flight Navigation

Who this is for: Anyone who knows high school trigonometry and vectors, but has ever wondered how a surveyor turns a tripod on a muddy hillside into legal property boundaries, or how an airplane pilot navigates across a continent without drifting off course in the wind.

1.1 Mapping a World That Refuses to Stand Still

If you took geometry in school, you learned to plot points on a neat Cartesian plane: x goes to the right, y goes up, and angles turn counterclockwise starting from the positive x-axis.

Then you step outside with a tripod, and the real world immediately breaks all three rules:

1. **North is up, not East.** In land surveying and aviation, 0° points straight toward North.
2. **Angles turn clockwise, like a clock face.** 90° is East, 180° is South, and 270° is West.
3. **The ground is not flat.** The earth is a bumpy, curved spheroid, and every measurement you make with a laser, tape, or compass accumulates physical error.

Centuries ago, when young surveyors like George Washington paced out Virginia valley tracts with a 66-foot iron chain, or when sea captains like Nathaniel Bowditch steered sailing ships by the stars, they faced the same fundamental challenge: **How do you keep small measurement errors from compounding into legal and physical disasters?**

If a boundary line is off by just a quarter of a degree, that error acts like a giant lever arm. Over a half-mile run, a property fence drifts by dozens of feet, overlapping a neighbor's land and sparking bitter legal disputes. In the air, a five-degree wind error over a three-hour flight can put an airplane fifty miles off course over fuel-empty wilderness.

The **StackCalc 35 (SC-35)** is designed specifically to tame this geometry. Instead of forcing you to balance parentheses or convert sexagesimal degrees in your head while standing in the rain or turbulence, it puts coordinate geometry (COGO), horizontal curve design, differential leveling, and wind-triangle vector calculations directly into a fast, transparent 4-level RPN stack.

1.1.1 Why Not Just Use a Phone App?

Modern surveyors have $30,000 robotic total stations, and pilots carry iPad tablets running GPS moving maps. Why does anyone want a dedicated pocket instrument?

- **Sunlight, Rain, and Gloves:** Capacitive phone screens wash out in harsh desert glare, register phantom touches in the rain, and refuse to work when you are wearing heavy leather work gloves or flight mittens. The SC-35 uses high-contrast reflective optics and crisp tactile mechanical keys you can feel through a glove.
- **The Legal Reality of Old Deeds:** Modern GPS gives coordinates in decimal degrees and 360° azimuths. But real property deeds recorded fifty or a hundred years ago are written in **Quadrant Bearings**: *"Thence South 42° 15′ 30″ West, 324.50 feet"*. Re-tracing a boundary requires typing quadrant angles directly.

- **Licensing Exam Rules:** The National Council of Examiners for Engineering and Surveying (NCEES) strictly bans smartphones, cellular tablets, and graphing calculators from Fundamentals of Surveying (FS) exams. Candidates must know how to solve traverse loops on a dedicated, approved instrument.

1.2 Core Concepts: How Surveyors and Pilots Think

1.2.1 Concept 1: The Compass Clock and Sexagesimal Angles

In high school math, you convert 125° 30′ into 125.5° before pressing [SIN]. But in surveying and navigation, angles are always read from optical theodolites, compass circles, and navigation charts in **Degrees, Minutes, and Seconds (*DDD° MM′ SS″*)**:

- $1^\circ = 60$ arcminutes ($'$)
- $1' = 60$ arcseconds ($''$)

On the SC-35, you don't need multi-key menu conversions. You enter sexagesimal angles as a single decimal number in canonical **DDD.MMSS** format:

- 125° 30′ 45″ is entered directly as **125.3045**.
- 45° 05′ 12″ is entered as **45.0512**.

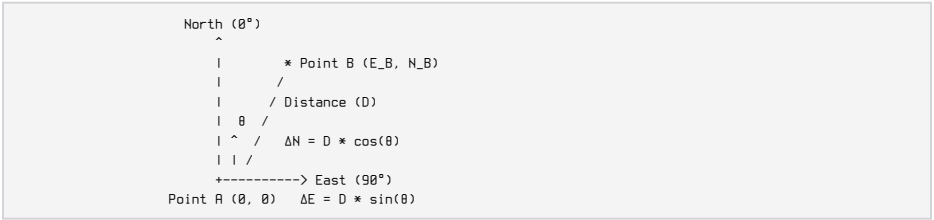
Softkey 5 (**DMS / DEG**) toggles the calculator between sexagesimal and decimal degrees instantly.

Why the Trig Functions Swap Places

Because surveying azimuths measure clockwise from North ($+y$) rather than counterclockwise from East ($+x$), the trigonometric coordinate displacements swap roles:

- **Departure (East displacement):** $\Delta E = \text{Distance} \cdot \sin(\text{Azimuth})$
- **Latitude (North displacement):** $\Delta N = \text{Distance} \cdot \cos(\text{Azimuth})$

If you enter an azimuth of 90° (due East), $\sin(90^\circ) = 1$ and $\cos(90^\circ) = 0$, so all your travel becomes East departure with zero North latitude.



1.2.2 Concept 2: The Two Fundamental Vector Problems (Forward & Inverse COGO)

Every land survey and aerial flight path comes down to two reciprocal questions:

1. **Forward COGO:** *"I am at point A. If I walk 500 feet along azimuth 045°, what are my new coordinates?"*

$$E_B = E_A + D\sin\theta, \quad N_B = N_A + D\cos\theta \tag{Eq. 1.1}$$

2. **Inverse COGO:** *"I have point A and point B on a map. What direction must I head, and how far must I travel to get from A to B?"*

$$\Delta E = E_B - E_A, \quad \Delta N = N_B - N_A \tag{Eq. 1.2}$$

$$\text{Distance} = \sqrt{\Delta E^2 + \Delta N^2}, \quad \text{Azimuth} = \operatorname{atan2}(\Delta E, \Delta N) \tag{Eq. 1.3}$$

On the SC-35, pressing [FWD] takes coordinates and vectors and outputs new coordinates; pressing [INV] takes two coordinate pairs and outputs distance and bearing in one keystroke.

1.2.3 Concept 3: Traverse Misclosure and the Shoelace Formula

When a survey crew walks around a parcel of land and returns to the starting benchmark, the sum of all North departures ($\sum \Delta N$) and East departures ($\sum \Delta E$) should theoretically equal zero. In practice, slight optical and taping tolerances leave a small gap: the **linear error of closure** (c):

$$c = \sqrt{(\sum \Delta E)^2 + (\sum \Delta N)^2} \tag{Eq. 1.4}$$

If the crew walked 2,500 feet in total perimeter, and the closure error is 0.15 feet, the precision ratio is:

$$\text{Precision} = \frac{2500}{0.15} \approx 16{,}667 \Rightarrow \mathbf{1:16{,}667} \tag{Eq. 1.5}$$

This exceeds the standard 1:10,000 legal code requirement for suburban boundary surveys.

To find the enclosed acreage, the SC-35 applies **Gauss's Shoelace Formula** across the sequential coordinates as they are entered into the $\Sigma+$ register:

$$\text{Area} = \frac{1}{2}\left| \sum_{i=1}^{n-1} (E_i N_{i+1} - E_{i+1} N_i) + (E_n N_1 - E_1 N_n) \right| \tag{Eq. 1.6}$$

Dividing square feet by 43,560 yields acres.

1.2.4 Concept 4: The Wind Triangle and Ground Speed

When an airplane flies through the atmosphere, the air mass itself is moving over the ground. The aircraft's actual path across the countryside (**Ground Track**) is the vector sum of its **True Airspeed Vector** ($\boldsymbol{V}_{\text{TAS}}$) and the **Wind Vector** ($\boldsymbol{V}_{\text{wind}}$).



By applying the Law of Sines to the triangle:

- **Wind Correction Angle (WCA):** The angle the nose of the plane must be turned *into* the wind so the airplane doesn't drift sideways off course.

- **Ground Speed (GS):** The actual speed the plane covers ground over the trees and runways.

1.3 Real-World Field Walkthroughs

1.3.1 Field Case 1: Checking a Disputed Rural Boundary (r/Surveying)

A land surveyor is verifying a boundary line between two iron boundary monuments.

- Benchmark 1 is at coordinates: 1000.00 N, 1000.00 E.
- Benchmark 2 is at coordinates: 1385.20 N, 1412.60 E.

What is the exact boundary distance and quadrant bearing between the monuments?

How to Solve It on the SC-35

1. Compute $\Delta E = 1412.60 - 1000.00 = 412.60$ ft.
2. Compute $\Delta N = 1385.20 - 1000.00 = 385.20$ ft.
3. Enter into the RPN stack:
   - **412.60** [ENTER] **385.20**
4. Tap [↗] [POL] (Rectangular to Polar / Inverse COGO):
   - Register X displays: **564.4842** (Distance in feet).
   - Register Y displays: **46.9745** (Azimuth in decimal degrees: 46.97°).
5. Convert decimal degrees to sexagesimal:
   - Tap [↖] [▸HR] → Register Y shows **46.5828** (46° 58′ 28″).
6. Press [BRG] then [↗] [SHOW]:
   - The LCD transiently displays: **N 46° 58' 28" E** (The exact quadrant bearing to write on the legal map).

1.3.2 Field Case 2: Flying with a 25-Knot Crosswind (r/flying)

You are cruising in a light aircraft at a True Airspeed of 120 knots heading on a True Course of 070°. Flight Service reports winds aloft at 120° at 25 knots. What heading should you fly, and what is your actual ground speed?

Step-by-Step Mechanics

1. **Find the Wind Angle:** 120° - 070° = 50° (wind is blowing from 50° to your right).
2. **Find the Crosswind Component:**
   - **25** [ENTER] **50** [SIN] [×] → **19.15 knots**.

3. **Find the Headwind Component:**
   - **25** [ENTER] **50** [COS] [×] → **16.07 knots**.
4. **Compute the Wind Correction Angle (WCA):**
   - $\sin(WCA) = 19.15/120 = 0.1596$
   - **19.15** [ENTER] **120** [÷] [↗] [ASIN] → **+9.18°**.
   - You must turn 9.2° right into the wind: **Heading = 079.2°**.
5. **Compute Actual Ground Speed:**
   - $GS = 120 \cdot \cos(9.18°)$ - 16.07 = 118.46 - 16.07 = **102.39 knots**.

1.4 Try It Yourself: Practice Problems

Problem 1: Circular Highway Curve Tangent

A civil engineering plan calls for a highway curve with an intersection deflection angle $\Delta = 36° 30′ 00″$ (36.5°) and a centerline radius $R = 1{,}200.00$ ft. Calculate the tangent distance $T = R \tan(\Delta / 2)$. *(Solution: $\Delta / 2 = 18.25°$. $1200 \times \tan(18.25°) =$* ***395.73 ft****.)*

Problem 2: Differential Leveling Loop

At setup 1, your optical level sights a rod on Bench Mark 1 (Elevation = 842.15 ft) with a backsight (*BS*) of 5.32 ft. The foresight (*FS*) to Turning Point 1 is 6.88 ft. What is the height of instrument (*HI*) and the elevation of Turning Point 1? *(Solution: HI = 842.15 + 5.32 =* ***847.47 ft****. Elevation = 847.47 - 6.88 =* ***840.59 ft****.)*

Problem 3: Density Altitude Check

An airport is at 3,500 ft elevation. On a hot summer day, ambient temperature is 35°C (95°F) with an altimeter setting of 29.72 inHg. Standard pressure altitude $PA = 3500 + (29.92 - 29.72) \times 1000 = 3{,}700$ ft. Standard temperature at 3,700 ft is $15 - 2 \times 3.7 = 7.6$°C. Calculate Density Altitude: $DA = PA + 120 \times (T_{\text{actual}} - T_{\text{standard}})$. *(Solution: DA = 3700 + 120 × (35 - 7.6) =* ***6,988 ft****. The airplane will perform on takeoff as if it were nearly 7,000 feet up!)*

SC-14 Industrial Trades, Electrical & Mechanical Handbook

Who this is for: Anyone who knows Ohm's Law ($V = IR$) and basic right-triangle trigonometry, but wants to understand the hidden physics behind electrical breaker panels, industrial motors, and bending steel pipes around structural obstacles.

2.1 When Ohm's Law Meets Heavy Copper

Have you ever noticed the lights in your home dim for a split second when the air conditioner compressor kicks on?

That momentary dip is Ohm's Law playing out along fifty feet of copper hidden behind your drywall. A copper wire is not an imaginary, zero-resistance superconductor; it is a very long, very thin resistor. When a heavy electric motor draws a sudden surge of current, that current passing through the wire's physical resistance drops voltage right in the line. The voltage at the appliance sags, and the missing energy turns directly into heat inside the wall.

In a home, an undersized wire might merely cause flickering lamps or a tripped breaker. But in a commercial hospital, a high-rise data center, or an industrial chemical plant, excessive electrical heat can melt insulation and cause electrical fires. To prevent this, the National Fire Protection Association publishes **NFPA 70**, known universally as the **National Electrical Code (NEC)**.

The NEC is essentially applied physics and geometry codified into law. Electricians, pipefitters, and precision machinists work in an environment where math must be calculated quickly, accurately, and without paper:

- Sizing a feeder cable so that a 480-volt industrial pump doesn't suffer brownout over a 300-foot run.
- Calculating how much steel conduit shrinks when bending around an overhead structural I-beam.
- Figuring out the exact spindle RPM for a carbide milling cutter so it cuts clean metal without overheating and dulling.

The **StackCalc 14 (SC-14)** puts these trade formulas, conductor tables, and mechanical calculations directly onto a rugged 4-level RPN stack.

2.1.1 Why Not Just Use a Smartphone?

- **The Scissor Lift Environment:** Picture standing on a scissor lift platform thirty feet in the air, wearing leather work gloves coated in greasy wire-pulling lubricant. A glass smartphone requires bare hands, risks cracking if dropped onto concrete, and turns off its screen mid-calculation. The SC-14 is built for one-handed operation with tactile mechanical switches that click clearly even through thick gloves.
- **Scrap Conduit is Expensive:** A single 10-foot stick of 3-inch or 4-inch rigid metal pipe costs over $150. If you miscalculate the offset travel or forget the shrinkage when bending around an obstacle, the pipe dog-legs, misses the junction box, and has to be thrown in the scrap bin.

- **Licensing Exams:** Journeyman and Master Electrician licensing exams strictly ban smartphones and internet-connected devices. Candidates must be able to calculate conductor ampacity, conduit fill, and three-phase motor loads on an approved standalone instrument.

2.2 Core Concepts: How Electricians and Machinists Think

2.2.1 Concept 1: The Vanishing Volts & Circular Mils (CM)

Why do electricians measure wire thickness in "Circular Mils" rather than square millimeters or inches?

A mil is one-thousandth of an inch (0.001 in). In standard geometry, the area of a circle is $(\pi/4)\ d^2$. But when sizing thousands of cables, carrying $\pi/4$ through every calculation is tedious.

Electrical pioneers invented a brilliant shortcut: **A Circular Mil (CM) is simply the diameter in mils, squared**:

$$CM = (d_{\text{mils}})^2 = (d_{\text{inches}} \times 1000)^2 \quad \textbf{(Eq. 2.1)}$$

A wire with a diameter of 0.10 inches is 100 mils thick, which means its area is simply 100^2 = 10,000 CM. The circle's π factor is baked directly into the material resistivity constant K:

- **Copper (K_{Cu}):** ≈ 12.9 Ω·cmil/ft at normal operating temperatures.
- **Aluminum (K_{Al}):** ≈ 21.2 Ω·cmil/ft.

The Voltage Drop Formula

For a single-phase two-wire circuit carrying current I over a one-way distance L (total wire loop length is $2L$):

$$\text{Voltage Drop } (VD) = \frac{2 \cdot K \cdot I \cdot L}{CM} \quad \textbf{(Eq. 2.2)}$$

The NEC recommends that voltage drop never exceed **3%** on a branch circuit (e.g. no more than 3.6 V drop on a 120 V line). If the calculated drop is too high, you must step up to a thicker wire gauge with more circular mils.

2.2.2 Concept 2: The Power of Three ($\sqrt{3} \approx 1.732$)

Why do commercial buildings and factories run on Three-Phase AC rather than the single-phase AC in our homes?

In single-phase power, voltage pulses like a heartbeat: it peaks, drops through zero, and reverses 60 times a second. A single-phase motor vibrates and needs auxiliary starting capacitors.

In three-phase power, three independent alternating voltage waves travel together, staggered evenly 120° apart. When one wave is passing through zero, the other two are delivering strong power. The total electrical power delivered is continuous and smooth, allowing motors to be smaller, lighter, and self-starting.

Because the three waves are separated by 120° on a circle, the line-to-line voltage is not simply double: it is multiplied by the geometric factor **√3 ≈ 1.732**:

$$V_{\text{line-to-line}} = \sqrt{3} \cdot V_{\text{phase-to-neutral}} \approx 1.732 \times 120\ \text{V} \approx 208\ \text{V} \quad \textbf{(Eq. 2.3)}$$

$$1.732 \times 277\ \text{V} \approx 480\ \text{V} \quad \textbf{(Eq. 2.4)}$$

Sizing Three-Phase Motor Circuits

For a 3-phase motor delivering horsepower (1 HP = 746 Watts) at efficiency η and power factor PF:

$$\text{Full Load Amps } (FLA) = \frac{\text{HP} \cdot 746}{\sqrt{3} \cdot V_{\text{line}} \cdot \eta \cdot PF} \quad \textbf{(Eq. 2.5)}$$

Under NEC Article 430, because electric motors draw a heavy inductive inrush current when starting up, conductors must be sized to **125% of motor FLA**. On the SC-14, pressing [↰] [3PH÷] automatically incorporates √3 in one keystroke.

2.2.3 Concept 3: Bending Steel Pipe with Right Triangles

When routing conduit across a ceiling, electricians frequently hit obstacles: air ducts, fire sprinkler lines, or structural steel columns. To get around them, you bend the pipe out and back in an **offset bend**:



An offset is simply two equal and opposite right triangles:

- **Travel Distance (Hypotenuse):** Travel = Depth · csc ϑ = Depth / sin ϑ
- **Conduit Shrinkage:** Because the hypotenuse is longer than the straight horizontal path, bending an offset pulls the end of the pipe backward toward the bender. If you don't account for this "shrinkage," your pipe will end up too short to reach the next junction box!

$$\text{Shrinkage} = \text{Depth} \times \left(\frac{1}{\tan\theta} - \frac{1}{\sin\theta}\right) \quad \textbf{(Eq. 2.6)}$$

Bending Multipliers Quick Reference

- **30° Bend:** Multiplier = 2.0 (Travel is exactly twice the depth; shrinks 1/4 in per inch of depth).
- **45° Bend:** Multiplier = 1.414 (Travel is 1.414 × Depth; shrinks 3/8 in per inch of depth).

2.2.4 Concept 4: Machining Feeds and Speeds

In a machine shop, cutting metal on a milling machine or lathe obeys a simple physical rule: the cutting tool must slide through the workpiece at a specific linear velocity called **Surface Feet per Minute (*SFM*)**:

- Mild steel cuts at roughly 100 SFM with high-speed steel.
- Aluminum can be machined at 600 SFM or higher.

To turn linear surface feet into rotational spindle speed (RPM):

$$RPM = \frac{SFM \times 12}{\pi \cdot \text{Diameter}} \approx \frac{3.82 \cdot SFM}{\text{Diameter (inches)}} \quad \textbf{(Eq. 2.7)}$$

A tiny 1/8-inch drill bit must spin at high RPM to achieve the proper cutting speed, while a massive 3-inch face mill must turn much slower to avoid overheating its carbide inserts.

2.3 Real-World Field Walkthroughs

2.3.1 Field Case 1: Sizing a 480V Sub-Panel Feeder (r/electricians)

An electrician is running a continuous 65-amp balanced 3-phase load from a main switchboard to a sub-panel located 220 feet away. The supply voltage is 480 V 3-phase copper.

1. What minimum ampacity wire is required under the NEC 125% continuous load rule?
2. If using #4 AWG copper wire (41,740 CM), what is the expected voltage drop and percentage drop?

How to Solve It on the SC-14

1. **Apply the 125% Continuous Rule (NEC 210.20):**
   - **65 [ENTER] 1.25 [×] → 81.25 Amps.**
   - Looking at NEC Table 310.16 via [TBL]: #4 AWG copper (rated for 85 A at 75°C) is required.
2. **Calculate 3-Phase Voltage Drop:**
   - Formula: $VD = \sqrt{3} \cdot K \cdot I \cdot L \;/\; CM$ ($K = 12.9$ for Cu, $I = 65$ A, $L = 220$ ft, $CM = 41740$).
   - **1.732 [ENTER] 12.9 [×] 65 [×] 220 [×]**
   - Stack displays: **319485**
   - **41740 [÷] → 7.65 Volts.**
3. **Calculate Percentage Drop:**
   - **7.65 [ENTER] 480 [÷] 100 [×] → 1.59%.**
   - Since $1.59\% < 3.0\%$, the wire easily satisfies code requirements without excessive heat.

2.3.2 Field Case 2: Bending a 4-Inch Offset Around an I-Beam (Practical Machinist)

An electrician running 1-inch EMT conduit must clear a steel ceiling beam that hangs 4 inches lower than the roof deck. The electrician chooses a standard 30° offset.

1. How far apart should the two bend marks be made on the pipe (Travel)?
2. How much will the pipe shrink toward the bender?

Step-by-Step Mechanics

1. **Compute Travel Distance:**
   - Multiplier for 30° is 1 / sin(30°) = 2.0.
   - **4** [ENTER] **30** [SIN] [÷] → **8.00 inches**.
   - The electrician places the two bend arrows exactly 8 inches apart on the conduit.
2. **Compute Pipe Shrinkage:**
   - For 30°, shrink is 1/4 inch per inch of offset depth.
   - **4** [ENTER] **0.25** [×] → **1.00 inch**.
   - The electrician shifts the first bend mark forward by 1 inch so the final end lands flush in the junction box.

2.4 Try It Yourself: Practice Problems

Problem 1: Motor Full Load Current

A three-phase 480-volt induction motor produces 25 HP on a factory conveyor line. The motor nameplate lists an operating efficiency of 88% ($\eta = 0.88$) and a power factor of 0.85 ($PF = 0.85$). Calculate the motor Full Load Amps (FLA). *(Solution: 25 × 746 = 18,650 W. Denominator = 1.732 × 480 × 0.88 × 0.85 = 622.3. FLA = 18650 / 622.3 =* ***29.97 Amps****. Branch conductors must be sized to 125% × 29.97 =* ***37.5 A****.)*

Problem 2: Conduit Raceway Fill

Under NEC Chapter 9 Table 1, a conduit raceway with more than two conductors may be filled to a maximum of **40%** of its cross-sectional area. A 1-inch EMT conduit has an internal cross-sectional area of 0.864 sq in. If each #10 THHN conductor has a cross-sectional area of 0.0211 sq in, what is the maximum number of #10 wires allowed in this pipe? *(Solution: Usable area = 0.864 × 0.40 = 0.3456 sq in. Max wires = 0.3456 / 0.0211 = 16.38* ⇒ ***16 wires****.)*

Problem 3: CNC Spindle Speed

A machinist is facing an aluminum block ($SFM = 500$) using a 2.5-inch diameter carbide face mill. Calculate the proper spindle speed in RPM. *(Solution: RPM = 3.82 × 500 / 2.5 =* ***764 RPM****.)*

SC-16 Systems Programming, Embedded & Cyber Operations

Who this is for: Anyone who writes software in high-level languages like Python, C, Rust, or JavaScript, but wants to understand what actually happens when variables turn into electrical signals, registers, and memory addresses inside a CPU.

3.1 When Numbers Hit the Silicon Ceiling

In high school math, numbers are smooth and infinite. You can start at zero and count upward forever, or take any two numbers and divide them into smaller and smaller fractions.

In a computer chip, numbers are neither smooth nor infinite. **They live in fixed-width boxes of silicon.**

A physical register in a microprocessor is made of microscopic flip-flops—transistor circuits that can hold either high voltage (1) or low voltage (0). An 8-bit register has eight switches; a 32-bit register has thirty-two; a 64-bit register has sixty-four. Because there are only so many switches in the box, numbers in computing behave like an old mechanical car odometer.

If your car's odometer has five digit wheels and you drive past 99,999 miles, the odometer doesn't expand to six digits; it clicks over to 00,000.

In 1996, the European Space Agency learned this lesson the hard way during the maiden launch of the Ariane 5 rocket. A routine software function tried to cram a 64-bit floating-point measurement of horizontal drift velocity into a 16-bit signed integer register. In pure math, the measured velocity of 32,768 was completely valid. But the maximum positive number that fits into a 16-bit signed integer is 32,767. The number overflowed, the CPU threw an unhandled hardware exception, the rocket's flight computers misread diagnostic dump data as an emergency command, and the rocket tore itself apart thirty-seven seconds into flight.

In the 1980s, Hewlett-Packard built the beloved **HP-16C Computer Scientist** to help programmers see and master these discrete boundaries. The **StackCalc 16 (SC-16)** modernizes that workflow: it lets you switch between Hex, Decimal, Octal, and Binary at a single touch, configure word sizes from 8 to 64 bits, mask bitfields, inspect compiler struct padding, and decode IEEE-754 floating-point bits directly on an RPN stack.

3.1.1 Why Not Just Use Python or an On-Screen Calculator?

- **The SCIF and Air-Gap Security Reality:** In defense aerospace, cybersecurity red-teams, and secure government facilities (governed by DoD ICD 705 standards), smartphones, fitness trackers, and internet-connected devices are strictly prohibited and locked outside the vault door. Engineers debugging secure crypto chips or flight avionics work in clean rooms with zero web access. The SC-16 provides a compliant, non-communicating physical instrument permitted inside secure areas.
- **Bare-Metal Hardware Bring-Up:** When a new prototype circuit board comes back from the factory, the microcontroller often crashes before the operating system or USB serial port even initializes. An engineer staring at raw hexadecimal addresses on a logic analyzer or JTAG debugger needs dedicated [A] through [↖] keys to mask bits and

calculate offsets without switching away from the oscilloscope.

- **Thinking in Hexadecimal:** Generic calculators force you to convert hex addresses into decimal before doing math. On the SC-16, you add **0x1000** to **0x0F80** and bit-shift directly in native hexadecimal.

3.2 Core Concepts: How Systems and Cyber Engineers Think

3.2.1 Concept 1: The Odometer in Reverse (Two's Complement)

How does a computer represent negative numbers using only ones and zeros?

Early computer pioneers tried using the first bit as a simple plus/minus flag. But this created two separate zeros (+0 and -0), requiring complicated arithmetic circuits.

The industry settled on a clever mechanical trick called **Two's Complement**. Think back to our five-digit car odometer:

- If you start at 00,000 and drive forward 1 mile, the odometer reads 00,001.
- What happens if you put the car in reverse and drive backward 1 mile? The wheels click backward from zero to **99,999**!

In a 16-bit binary register:

- 0 is **0000 0000 0000 0000**.
- Rolling backward by 1 gives **1111 1111 1111 1111** (65,535 if unsigned, or **-1** if signed!).

To negate any binary number in two's complement, you simply **invert all the bits and add 1**:

$$\text{Negate}(x) = (\sim x) + 1 \quad \textbf{(Eq. 3.1)}$$

Because of this symmetry, a CPU does not need a separate subtraction circuit. To compute A - B, the silicon simply computes A + ((~ B) + 1) using standard addition gates.

Word Sizes and the Sign Bit

The most significant (leftmost) bit acts as the sign flag:

- **0** means positive.
- **1** means negative.

Word Size	Bit Width	Unsigned Range	Signed Two's Complement Range
BYTE	8 bits	0 to 255	-128 to +127
WORD	16 bits	0 to 65,535	-32,768 to +32,767
DWORD	32 bits	0 to 4,294,967,295	-2,147,483,648 to +2,147,483,647
QWORD	64 bits	0 to ≈ 1.84×10^{19}	≈ -9.22×10^{18} to $+9.22 \times 10^{18}$

3.2.2 Concept 2: Speaking in Hexadecimal (Base-16)

Humans have ten fingers, so we count in decimal (base-10). Computers have binary switches, so they count in binary (base-2).

Writing long binary strings like **1101111010101101** is dizzying and prone to transcription errors. Hexadecimal (base-16) solves this because $16 = 2^4$. **Every single hexadecimal digit represents exactly four binary bits (a "nibble")**:

Hex Digit	Binary	Decimal	Hex Digit	Binary	Decimal
0–9	**0000–1001**	0–9	**C**	**1100**	12
A	**1010**	10	**D**	**1101**	13
B	**1011**	11	**E / F**	**1110 / 1111**	14 / 15

The 16-bit binary number **1101 1110 1010 1101** splits cleanly into four nibbles: **DE AD ⇒ 0xDEAD**.

3.2.3 Concept 3: Bitwise Logic & Masking

In high-level programming, you write conditions like **if (user.isActive)**. In firmware and network packet drivers, you interact directly with hardware registers where each individual bit controls a physical circuit (like turning on an LED, enabling an antenna, or resetting a timer).

We control individual bits using Boolean logic:

- **AND (&):** Clears bits to zero. To check if bit 3 is set, you "mask" with **0000 1000**.
- **OR (|):** Sets bits to one without touching other bits.
- **XOR (^):** Toggles bits, or checks if two values differ.
- **Bit Shifts (<<, >>):** Multiplying or dividing by powers of 2. Shifting left by 1 multiplies by 2; shifting right by 1 divides by 2.

3.2.4 Concept 4: Memory Alignment & Struct Padding

Modern 32-bit and 64-bit processors do not read memory one single byte at a time; they fetch 4-byte or 8-byte chunks across wide silicon buses.

If an 8-byte variable (like a 64-bit float) sits at an odd memory address (like **0x1001**), the CPU has to make two separate memory accesses, stitch the halves together, and burn clock cycles. To prevent this, compilers insert invisible **padding bytes** between struct variables:

```
struct SensorData {
    uint8_t  sensorId;   // 1 byte  at offset 0
    // [3 PADDING BYTES INSERTED BY COMPILER]
    uint32_t timestamp;  // 4 bytes at offset 4 (must align to
        multiple of 4!)
    uint16_t reading;    // 2 bytes at offset 8
    // [2 PADDING BYTES INSERTED AT END]
}; // Total size = 12 bytes, NOT 7 bytes!
```

The SC-16's **ALIGN** softkeys calculate struct offsets and padding gaps automatically.

3.3 Real-World Field Walkthroughs

3.3.1 Field Case 1: Isolating a Hardware Status Bit (r/embedded)

An engineer is reading an SPI status register from a radio transceiver. The hardware datasheet says:

- Register address: **0x4001**

- Bit 5 is the **TX_READY** flag (1 = ready to transmit, 0 = busy).
- The microcontroller reads raw byte: **0x7A**.

Is the radio ready to transmit?

How to Solve It on the SC-16

1. Put the SC-16 in **HEX** mode: softkey [HEX].
2. Enter the raw byte: **7A** [ENTER].
3. Create a bitmask for bit 5 ($2^5 = 32 = 0x20$):
   - **20**
4. Execute bitwise AND:
   - Tap [↶] [AND].
   - Register *X* displays: **0x20**.
   - Because the result is non-zero (**0x20**), bit 5 is high: **The radio is ready to transmit!**

3.3.2 Field Case 2: Subnetting an IPv4 Network (Hacker News)

A network architect is dividing a corporate network with IP block **192.168.10.0/26**.

1. What is the subnet mask in dotted-decimal and hex?
2. How many total host IP addresses are available in each subnet?

Step-by-Step Mechanics

1. A /**26** prefix means the first 26 bits are 1s, and the remaining 32 - 26 = 6 bits are 0s.
2. In the 4th byte, there are two 1s followed by six 0s: **1100 0000** in binary.
   - On the SC-16, type **11000000** in **BIN** mode → switch to **DEC**: displays **192**.
   - Subnet mask is **255.255.255.192** (or **0xFFFFFFC0** in hex).
3. Compute total host count from the 6 host bits:
   - **2** [ENTER] **6** [y^x] → **64 total addresses**.
   - Subtract network ID (all 0s) and broadcast ID (all 1s): 64 - 2 = **62 usable hosts**.

3.4 Try It Yourself: Practice Problems

Problem 1: Two's Complement Negative Conversion

Using an 8-bit word size (**BYTE** mode on SC-16), enter decimal value +42. Convert it to signed negative -42 using two's complement. *(Solution: +42 = 0x2A = 0010 1010₂. Invert bits: 1101 0101₂ = 0xD5. Add 1: 0xD5 + 1 = **0xD6**. In 8-bit signed decimal, 0xD6 = **-42**.)*

Problem 2: Circular Ring-Buffer Masking

A high-speed UART driver allocates a ring buffer of size 256 bytes (2^8). The head pointer is currently at index **310**. Using bitwise AND masking instead of an expensive division/modulo operator (**%**), find the buffer index. *(Solution: Mask with size minus one: 256 - 1 = 255 = 0xFF.* **310 & 0xFF ⇒ 310** [ENTER] **255** [AND] → **54**. *Fast, single-cycle ring-buffer wrapping!)*

Problem 3: Decoding an IEEE-754 Single-Precision Float

A binary packet dumps the 32-bit hex word **0x41440000**. On the SC-16, enter **41440000** in **HEX** mode, then tap [FLT] followed by [↗] [SHOW]. *(Solution: Sign bit = 0 (+). Exponent bits = 10000010_2 = 130. Unbias: 130 - 127 = 3. Mantissa bits = 0.28125. Value = $+1.28125 \times 2^3$ =* ***12.25****.)*

SC-24 Cinema Optics, Broadcast Timecode & Sound Handbook

Who this is for: Anyone who loves photography, filmmaking, or audio production and wants to understand the exact mathematical physics connecting lenses, camera sensors, decibels, and broadcast video sync.

4.1 When Light and Sound Become Numbers

Every motion picture you have ever watched is an optical illusion. A projector or screen flashes twenty-four still photographs in front of your eyes every second. Because of the human brain's persistence of vision, our visual cortex fuses those discrete snapshots into the seamless illusion of continuous motion.

Our ears, by contrast, are far more sensitive to time. While our eyes are easily fooled by 24 frames a second, our hearing can detect an acoustic echo or phase delay of just a few milliseconds.

On a professional movie set or live broadcast production, these two worlds of physics—photons hitting camera sensors and sound waves striking microphones—must be synchronized down to the microsecond. A film crew operates under relentless pressure:

- When the sun is setting during "magic hour," the light drops by an entire exposure stop every five minutes.
- If a camera operator chooses the wrong f-stop, an actor leaning forward by two inches blurs completely out of focus.
- If a live sports broadcast drifts by a few frames, audio loses lip-sync, and the transmission misses satellite commercial break windows.

The **StackCalc 24 (SC-24)** puts frame-accurate SMPTE timecode arithmetic, optical depth-of-field geometry, Exposure Value (EV) lighting photometrics, and acoustic decibel summing onto a fast, dedicated RPN stack.

4.1.1 Why Not Just Use a Phone App?

- **The Magic Hour Window:** When racing against fading daylight on an exterior film set, every minute costs thousands of dollars in crew wages. Unlocking a phone, squinting against glare, and tapping through menus wastes precious light. The SC-24 lets a 1st AC (Focus Puller) or Director of Photography calculate hyperfocal distances and neutral density filter factors in two tactile clicks.
- **Set Discipline & Silence:** Phone screens illuminate dark cinema sets and notifications distract actors during rehearsals. The SC-24 is a silent, non-radiating tool designed for professional working environments.
- **Frame-Accurate Arithmetic:** Standard calculators work in base-10 decimals, but film and broadcast video operate in mixed radices: 24 frames per second for film, 25 fps for European TV, and 29.97 fps for North American broadcast. Trying to add **01:14:22:18** to **00:45:10:12** on a decimal calculator requires endless manual conversions.

4.2 Core Concepts: How Cinematographers and Sound Engineers Think

4.2.1 Concept 1: The 0.1% Color Mystery (SMPTE Drop-Frame Timecode)

Timecode is the universal coordinate system of media:

$$\mathbf{HH:MM:SS:FF} \quad \text{(Hours : Minutes : Seconds : Frames)} \qquad \textbf{(Eq. 4.1)}$$

Why does North American television run at the bizarre rate of **29.97 frames per second** instead of an even 30.00?

In the 1940s, black-and-white television ran at exactly 30 fps. But in 1953, when color television was invented, engineers discovered that the new high-frequency color subcarrier (3.58 MHz) caused electrical interference with the audio carrier wave. To eliminate this interference, the broadcast industry decided to slow down the television frame rate by exactly **0.1%**:

$$\text{Color Framerate} = \frac{30}{1.001} \approx 29.97002997 \text{ fps} \qquad \textbf{(Eq. 4.2)}$$

This tiny 0.1% difference creates a giant headache:

- At 30 fps, 108,000 frames equal exactly 1 hour of wall-clock time.
- At 29.97 fps, 108,000 frames take **1 hour, 3 seconds, and 18 frames**!

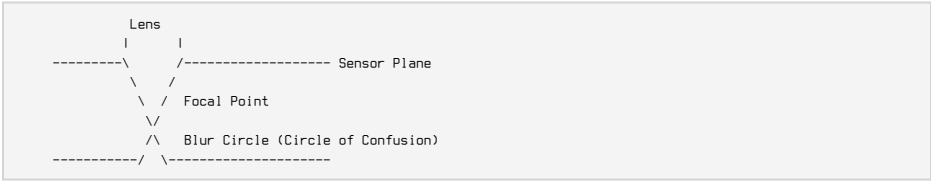
Over an hour of broadcast, video drifts nearly 3.6 seconds behind the wall clock. To fix this, the Society of Motion Picture and Television Engineers (SMPTE) created **Drop-Frame Timecode (DF)**:

- Every minute, the timecode counter **skips frame numbers 00 and 01**.
- **Exception:** To prevent over-correcting, it does *not* skip frames on minute numbers ending in zero (minutes 00, 10, 20, 30, 40, 50).

No actual video frames are deleted; only the numerical labels are dropped so the timecode display stays locked to real-world clocks. The SC-24 handles this drop-frame arithmetic automatically.

4.2.2 Concept 2: The Slippery Edge of Focus (Depth of Field & Hyperfocal Distance)

A camera lens can only bring one exact distance into sharp mathematical focus. Points closer or farther away project onto the camera sensor as small, blurred discs of light called **Circles of Confusion (*CoC*)**.



As long as the circle of confusion is smaller than the human eye can distinguish at a standard viewing distance (roughly 0.03 mm on a Full Frame sensor, or 0.02 mm on Super 35mm cinema), the image appears tack-sharp. The region where blur is imperceptible is the **Depth of Field (DOF)**.

Hyperfocal Distance (H)

If you focus a lens on infinity, half the light is wasted on stars, and nearby foreground objects blur.

If you instead focus at the **Hyperfocal Distance (H)**, everything from **half that distance ($H/2$) all the way to infinity** remains in sharp focus:

$$H \approx \frac{f^2}{N \cdot c} \quad \textbf{(Eq. 4.3)}$$

where f is focal length (in mm), N is f-stop number, and c is circle of confusion (in mm).

4.2.3 Concept 3: The Decibel Mystery & Acoustic Summation

Why is human hearing logarithmic rather than linear?

If you hear one person talking at a dinner table, and a second person speaks at the exact same volume, your ear does *not* perceive the room as twice as loud. To sound twice as loud to a human ear, sound energy must increase by roughly **ten times**!

Sound engineers measure acoustic pressure on a logarithmic decibel scale:

$$L_p = 20\log_{10}\left(\frac{P}{P_0}\right) \quad [\text{dBSPL}] \quad \textbf{(Eq. 4.4)}$$

where $P_0 = 20$ μPa is the quietest sound a human ear can detect.

Adding Two Sound Sources ($dB\oplus$)

If a guitar amplifier produces 90 dBSPL, and a second identical amplifier is turned on producing 90 dBSPL, what is the total volume?

You cannot add decibels directly ($90 + 90 \neq 180$). You must convert back to linear power, add the energies, and take the logarithm again:

$$L_\Sigma = 10\log_{10}(10^{90/10} + 10^{90/10}) = 10\log_{10}(2 \times 10^9) = 90 + 3.01 \approx \mathbf{93}\ \text{dBSPL} \quad \textbf{(Eq. 4.5)}$$

Two identical uncorrelated sound sources always add exactly +3 dB. The SC-24 features a dedicated [dB⊕] key that sums sound levels in one step.

4.3 Real-World Field Walkthroughs

4.3.1 Field Case 1: Editing a 2-Hour Live Concert (Cinematography.com)

A post-production editor is assembling two live camera reels recorded at 29.97 fps Drop-Frame:

- Reel 1 Duration: **01:14:28:12**
- Reel 2 Duration: **00:52:45:20**

What is the exact combined running time of the broadcast cut?

How to Solve It on the SC-24

1. Set framerate to 29.97 DF using softkey [TC].
2. Enter Reel 1: **01142812** [ENTER].
3. Enter Reel 2: **00524520** [+].
4. Register X displays: **02:07:14:02** (2 hours, 7 minutes, 14 seconds, and 2 frames).

The editor knows the exact master duration without manually tracking dropped frames.

4.3.2 Field Case 2: Deep-Focus Landscape Cinematography (r/cinematography)

A filmmaker is shooting an exterior scene with a 35mm lens on a Full Frame cinema sensor ($c = 0.030$ mm) at $f/8$.

1. What is the hyperfocal distance?
2. If the lens is focused at the hyperfocal distance, how close can an actor stand to the camera and still remain sharp?

Step-by-Step Mechanics

1. **Compute Hyperfocal Distance:**
   - Formula: $H = f^2 \,/\, (N \cdot c) = 35^2 \,/\, (8 \times 0.030) = 1225/0.24 = 5{,}104$ mm $\approx$ **5.10 meters** (16.7 feet).
   - On the SC-24: **35** [x^2] **8** [÷] **0.03** [÷] $\rightarrow$ **5104 mm**.
2. **Compute Minimum Sharp Distance:**
   - Near limit $= H \,/\, 2 = 5.10 \,/\, 2 =$ **2.55 meters** (8.4 feet).
   - Everything from 8.4 feet out to the distant mountains on the horizon will be in focus.

4.4 Try It Yourself: Practice Problems

Problem 1: Acoustic Decibel Summing

At a live concert, three uncorrelated noise sources are measured at the soundboard:

- Main PA: 96 dBSPL
- Crowd cheering: 92 dBSPL
- Stage monitor bleed: 88 dBSPL

Calculate the total combined sound pressure level at the mixing console using [dB⊕]. *(Solution: $10^{9.6} + 10^{9.2} + 10^{8.8} = 3.981 \times 10^9 + 1.585 \times 10^9 + 0.631 \times 10^9 = 6.197 \times 10^9$. $10 \log_{10}(6.197 \times 10^9) =$ **97.92 dBSPL**.)*

Problem 2: Neutral Density (ND) Filter Selection

A cinematographer is filming outdoors at ISO 100 with a shutter speed of 1/50 second. A light meter reading indicates an Exposure Value of $EV = 15$ (which requires $f/16$ for proper exposure). However, the director wants a shallow depth of field at $\boldsymbol{f/2.0}$ (an increase of 6 full stops of light: $f/16 \rightarrow 11 \rightarrow 8 \rightarrow 5.6 \rightarrow 4 \rightarrow 2.8 \rightarrow 2$). What optical density (OD) neutral density filter is required to cut 6 stops of light? *(Solution: Each stop is 0.3 OD. 6 stops × 0.3 =* ***1.8 ND*** *(an ND64 filter, transmitting only 1/64 = 1.56% of light).)*

Problem 3: Sound Delay Tower Alignment

At an outdoor music festival, a secondary speaker delay tower is positioned 175 feet away from the main stage. The air temperature is 20°C (speed of sound $c \approx 1{,}125$ ft/s). Calculate the required delay in milliseconds to sync the speakers. *(Solution: Time = 175 ft / 1125 ft/s = 0.1555 seconds =* ***155.6 milliseconds****.)*

SC-50 RF Spectrum, Transmission Lines & Microwave Handbook

Who this is for: Anyone who understands complex numbers ($a + jb$) and alternating current, but wants to know why high-frequency radio waves behave like sound echoes bouncing off canyon walls, and how engineers tune antennas to prevent expensive amplifiers from melting.

5.1 When Wires Stop Acting Like Wires

If you shout into a wide canyon, your voice travels outward through the air until it hits the opposite cliff face. Because solid granite has a vastly different acoustic density than thin air, the rock cannot absorb the sudden wave of kinetic energy. The sound wave bounces straight back as an audible echo.

At normal household frequencies (60 Hz AC), electrical current travels along a copper wire smoothly and uniformly. The electromagnetic wavelength at 60 Hz is over 3,000 miles long—far larger than your house—so voltage is virtually identical everywhere along the cord.

Now crank the frequency up to 1 Gigahertz (1,000,000,000 cycles per second), the realm of Wi-Fi, cellular 5G, and aircraft radar.

At 1 GHz, the wavelength shrinks to just **30 centimeters** (about one foot). A coaxial cable running from a radio transmitter to an antenna on a tower is no longer a simple static connection: it is a **distributed transmission line**. The electrical signal travels along the wire as an electromagnetic wave.

When that forward-traveling wave hits the antenna terminals, one of two things happens:

1. If the antenna's electrical impedance **matches** the cable, the wave steps cleanly into the air and radiates into space as radio waves.
2. If the impedance **does not match**, the antenna acts like the granite cliff in our canyon: it cannot absorb the energy, and a portion of the wave **bounces backward down the cable toward the transmitter**.

That reflected wave collides with the incoming wave, creating standing hot spots of voltage along the feedline. Worse, when that reflected power hits the delicate output transistors of the transmitter, it is dissipated as instant heat. In a 100-watt or 1,000-watt transmitter, a mismatched antenna can melt a $500 Gallium-Nitride (GaN) power transistor in milliseconds.

The **StackCalc 50 (SC-50)** is designed specifically to calculate these wave reflections. It pairs a native complex-number RPN stack with dedicated reflection engines, one-touch characteristic impedance toggles (50 Ω ↔ 75 Ω), Friis receiver noise figure accumulators, and transmission line matching tools.

5.1.1 Why Not Just Use a Phone App?

- **The Tower Climbing Reality:** When a broadcast or cellular technician is climbing a 150-foot radio tower in gusty winds, tethered to a safety harness, a delicate touchscreen smartphone is an accident waiting to happen. The SC-50 features tactile

keys that can be operated reliably with one hand while tethered to a climbing lanyard.

- **The 50 Ω vs. 75 Ω Compromise:** Why is the entire RF world built on 50 Ω and 75 Ω? In the 1930s, electrical researchers discovered that for coaxial cables with air insulation, minimum signal loss (attenuation) occurs at 77 Ω, while maximum power-handling capability occurs at 30 Ω. Radio transmitters standardized on **50 Ω** as the sweet-spot compromise between power and loss. Cable television (CATV), which carries tiny microvolt signals over long neighborhood runs, standardized on **75 Ω** to minimize attenuation. The SC-50 lets you toggle between 50 Ω and 75 Ω reference systems with a single key.

5.2 Core Concepts: How RF and Microwave Engineers Think

5.2.1 Concept 1: The Bouncing Wave and Reflection Coefficient (Γ)

When a transmission line with characteristic impedance Z_0 (usually 50 Ω) connects to an antenna with complex load impedance $Z_L = R + jX$, the ratio of the reflected voltage wave to the forward voltage wave is the **Voltage Reflection Coefficient (Γ, Greek gamma)**:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \rho \angle \theta \qquad \textbf{(Eq. 5.1)}$$

Because Z_L is a complex number, Γ is also a complex phasor with a magnitude ($\rho = |\Gamma| \leq 1.0$) and a phase angle (ϑ):

- **Perfect Match ($Z_L = 50\ \Omega$):** Γ = (50 - 50) / (50 + 50) = **0**. Zero reflected wave; 100% of power radiates into the sky!
- **Open Circuit ($Z_L = \infty$):** Γ = **+1.0** ∠ 0°. All voltage reflects back in-phase.
- **Short Circuit ($Z_L = 0$):** Γ = **-1.0** = 1.0 ∠ 180°. All voltage reflects back inverted.

5.2.2 Concept 2: Standing Waves (VSWR) & Return Loss

When forward and reflected waves pass through each other inside a cable, their peaks and troughs interfere constructively and destructively, creating a stationary wave pattern.

The ratio of the maximum voltage peak to the minimum voltage valley is the **Voltage Standing Wave Ratio (VSWR)**:

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + \rho}{1 - \rho} \qquad \textbf{(Eq. 5.2)}$$

- **Ideal Match:** $VSWR$ = **1.0 : 1** (smooth, uniform voltage along the cable).
- **Good Commercial Antenna:** $VSWR \leq$ **1.5 : 1** (4% power reflected).
- **Dangerous Mismatch:** $VSWR \geq$ **3.0 : 1** (25% or more power reflected backward into the transmitter!).

Return Loss (RL)

Engineers often prefer to express reflection efficiency on a decibel scale called **Return Loss**:

$$RL = -20\log_{10}|\Gamma| \quad [\text{dB}] \qquad \textbf{(Eq. 5.3)}$$

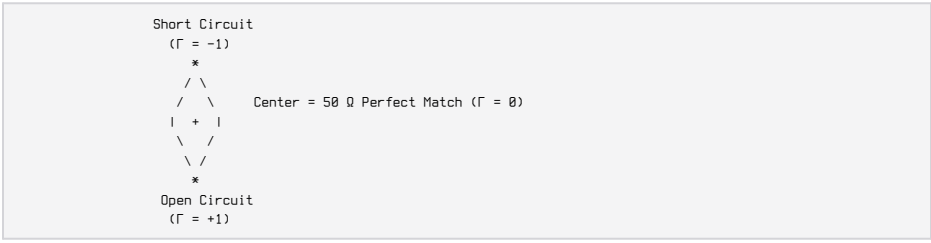
A high return loss means the reflected wave has been heavily attenuated:

- $RL = 20$ dB ⇔ $|\Gamma| = 0.10$ ⇔ 99% of power is absorbed by the antenna.
- $RL = 10$ dB ⇔ $|\Gamma| = 0.316$ ⇔ 90% of power is absorbed (10% bounces back).

5.2.3 Concept 3: Folding Infinity into a Circle (The Smith Chart)

In 1939, RCA engineer Phillip Smith grew tired of solving complex hyperbolic tangent equations to figure out how impedance changes as you travel along a transmission line.

He noticed something magical: if you map the infinite right half of the complex impedance plane ($R \geq 0$) onto the complex reflection coefficient Γ, **the entire infinite plane folds into a compact circular disk with a radius of 1.0**:



- The exact center of the circle is the **50 Ω perfect match** ($\Gamma = 0$).
- The outer boundary rim is total reflection ($|\Gamma| = 1.0$).
- Moving down a coaxial cable corresponds simply to rotating clockwise around the center of the chart!

On the SC-50, pressing [SMITH] lets you navigate impedance transformations and design matching networks without needing paper charts.

5.2.4 Concept 4: Listening to the Stars (Friis Cascaded Noise)

When a satellite dish points at deep space, the signal arriving from a spacecraft millions of miles away is unimaginably faint: often less than a billionth of a microwatt (-120 dBm).

Every electronic component generates natural thermal noise (random motion of electrons). To amplify a faint signal without burying it in hiss, engineers use a chain of amplifiers.

Harald Friis at Bell Labs discovered that the total noise figure (F) of an entire multi-stage receiver is dominated by the **very first stage**:

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots \qquad \textbf{(Eq. 5.4)}$$

Notice the denominator: any noise added by stage 2 (F_2) is divided by the gain of stage 1 (G_1)!

This explains why satellite dishes and cellular base stations always place a high-gain, Low Noise Amplifier (LNA) right at the antenna feedhorn on the mast, rather than down inside the warm building at the other end of a long, lossy cable.

5.3 Real-World Field Walkthroughs

5.3.1 Field Case 1: Checking an Amateur Radio Antenna Match (Microwaves101)

An antenna analyzer measures the feedpoint impedance of a newly installed dipole antenna at 14.2 MHz:

- Load impedance: $Z_L = 35 - j25\ \Omega$
- Reference line: $Z_0 = 50\ \Omega$

What is the reflection coefficient magnitude ($|\Gamma|$), the VSWR, and the Return Loss?

How to Solve It on the SC-50

1. Enter Z_L on the complex stack:
   - **35** [ENTER] **25** [+/−] (Stack holds 35 - $j25$).
2. Enter Z_0:
   - **50**
3. Tap [SWR]:
   - Register X displays: **1.884** (**VSWR = 1.88 : 1**).
   - Register Y displays: **0.3065** (Reflection coefficient $|\Gamma| = 0.307$).
4. Tap [↶] [RL]:
   - Register X displays: **10.27 dB** (Return Loss).
   - About 90.6% of the transmitter's power is radiating into the air, and 9.4% is reflected back. A simple L-matching network can tune this to 1.0 : 1.

5.3.2 Field Case 2: Satellite Downlink Low-Noise Preamplifier (EEVblog)

A satellite ground station receiver consists of:

- Stage 1 (Masthead LNA): Noise Figure = 0.8 dB ($F_1 = 1.202$), Gain = 20 dB ($G_1 = 100$).
- Stage 2 (Coax Cable Run): Loss = 3.0 dB ($F_2 = 2.0$, $G_2 = 0.50$).
- Stage 3 (Indoor Receiver): Noise Figure = 6.0 dB ($F_3 = 3.98$).

What is the total cascaded noise figure of the receiving system?

Step-by-Step Mechanics

1. **Apply Friis Equation:**

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} \quad \textbf{(Eq. 5.5)}$$

$$F_{\text{total}} = 1.202 + \frac{2.0 - 1.0}{100} + \frac{3.98 - 1.0}{100 \times 0.50} = 1.202 + 0.010 + 0.0596 = \mathbf{1.2716} \quad \textbf{(Eq. 5.6)}$$

2. **Convert Back to Decibels:**

$$NF_{\text{total}} = 10\log_{10}(1.2716) = \mathbf{1.04}\ \text{dB} \quad \textbf{(Eq. 5.7)}$$

Even though the indoor receiver has a noisy 6 dB noise figure and the cable is lossy, the masthead LNA's 20 dB gain keeps the total system noise figure down to an ultra-clean 1.04 dB!

5.4 Try It Yourself: Practice Problems

Problem 1: Transmission Line Quarter-Wave Transformer

A high-frequency load has a purely resistive impedance $Z_L = 100\ \Omega$. You need to match it to a 50 Ω coaxial generator using a quarter-wavelength ($\lambda / 4$) section of transmission line. Calculate the required characteristic impedance of the quarter-wave line: $Z_{\text{match}} = \sqrt{(Z_0 \cdot Z_L)}$. *(Solution: $Z_{match} = \sqrt{(50 \times 100)} = \sqrt{5000} \approx$ **70.71 Ω**.)*

Problem 2: Return Loss from Power Measurements

A forward-power directional coupler measures 100 Watts forward power toward an antenna and 4 Watts reflected power bouncing back.

1. What is the reflection coefficient magnitude $\rho = \sqrt{(P_{\text{ref}} / P_{\text{fwd}})}$?
2. What is the Return Loss ($RL = -20 \log_{10}\rho$)?
3. What is the VSWR?

*(Solution: $\rho = \sqrt{(4 / 100)} =$ **0.20**. $RL = -20 \log_{10}(0.20) =$ **13.98 dB**. $VSWR = (1 + 0.20) / (1 - 0.20) = 1.20 / 0.80 =$ **1.50 : 1**.)*

Problem 3: Resonant LC Tank Frequency

A resonant filter circuit consists of an inductor $L = 47$ nH (47×10^{-9} H) and a capacitor $C = 10$ pF (10×10^{-12} F). Calculate the resonant frequency $f_0 = 1 / 2\pi\sqrt{LC}$ in Megahertz. *(Solution: $LC = 47 \times 10^{-9} \times 10 \times 10^{-12} = 4.7 \times 10^{-19}$. $\sqrt{LC} \approx 6.8556 \times 10^{-10}$. $2\pi\sqrt{LC} \approx 4.3075 \times 10^{-9}$. $f_0 = 1 / 4.3075 \times 10^{-9} = 232{,}150{,}000$ Hz = **232.15 MHz**.)*

SC-60 Wet-Lab Chemistry, Stoichiometry & Spectroscopy Handbook

Who this is for: Anyone who has taken general chemistry and wants to understand how bench scientists, biochemists, and pharmaceutical researchers turn chemical formulas into exact recipes on a laboratory balance.

6.1 Counting What You Cannot See

Imagine you want to react one molecule of citric acid with three molecules of sodium bicarbonate.

The chemical equation looks tidy on paper:

$$C_6H_8O_7 + 3\,NaHCO_3 \rightarrow Na_3C_6H_5O_7 + 3\,H_2O + 3\,CO_2 \quad \textbf{(Eq. 6.1)}$$

The ratio is simple: 1 to 3. But when you walk into the laboratory, you face an immediate physical barrier: **molecules are far too tiny to see or count.** You cannot pick up individual molecules with micro-tweezers and drop them into a beaker. The only tool you have to measure chemicals on the lab bench is an analytical balance that measures mass in **grams**.

How do you translate an exact count of trillions of invisible molecules into a pile of white powder on a glass weighing dish?

The bridge between the atomic world and the human world is the **Mole**: a quantity defined by Avogadro's constant (6.022×10^{23} particles). By summing the atomic weights of each element on the periodic table, chemists calculate the **Molecular Weight (MW)** in grams per mole.

Once you leave the dry balance and dissolve powders into liquid beakers, chemistry becomes a world of concentrations, dilutions, and logarithmic pH scales. A researcher formulating a buffer or synthesizing a medicine must navigate:

- Calculating exact mass for target molarities ($m = C \cdot V \cdot MW$).
- Diluting concentrated chemical stocks ($C_1\ V_1 = C_2\ V_2$).
- Designing biochemical buffers that keep delicate proteins from denaturing.
- Measuring how much light a solution absorbs to determine its purity.

The **StackCalc 60 (SC-60)** puts an IUPAC molecular weight tokenizer, Henderson-Hasselbalch buffer solver, serial dilution calculator, and the Beer-Lambert spectrophotometry law directly into a chemical 4-level RPN stack.

6.1.1 Why Not Just Use a Phone App?

- **The Nitrile Glove and Chemical Hazard Barrier:** In an active chemistry or biology lab, researchers wear disposable nitrile gloves exposed to concentrated acids, organic solvents (like chloroform or DMSO), and mutagenic DNA stains (like ethidium bromide). Touching a personal glass smartphone with contaminated gloves violates laboratory biosafety standards and transfers toxic compounds to personal items and the home. The SC-60 is an IP-rated, solvent-resistant laboratory instrument designed to live

next to the balance and be wiped down with ethanol or bleach.

- **The Hydrate Mass Trap:** Many common laboratory salts absorb water molecules into their crystal lattice. For example, anhydrous copper sulfate ($CuSO_4$) has a molar mass of 159.61 g/mol, while copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) has a molar mass of 249.68 g/mol—a massive 56% difference! If a student reads a protocol calling for $CuSO_4$ and weighs the pentahydrate without compensating for the water weight, the solution is severely under-concentrated, ruining weeks of research. The SC-60's formula tokenizer computes exact hydrate stoichiometries directly on the screen.

6.2 Core Concepts: How Chemists Think

6.2.1 Concept 1: The Molecular Bridge (Molarity & Mass Prep)

A solution's concentration is measured in **Molarity (M)**: the number of moles of solute dissolved in one liter of total solution:

$$\text{Molarity } (C) = \frac{\text{Moles } (n)}{\text{Volume in Liters } (V)} \quad \left[\frac{\text{mol}}{\text{L}}\right] \tag{Eq. 6.2}$$

To prepare a solution of known concentration C and volume V, the mass of solid chemical powder to weigh on the balance is:

$$\text{Mass } (m) = C \cdot V \cdot MW \quad [\text{grams}] \tag{Eq. 6.3}$$

For example, to make 500 mL (0.50 L) of a 0.20 M sodium chloride (NaCl, $MW = 58.44$ g/mol) solution:

$$m = 0.20 \text{ mol/L} \times 0.50 \text{ L} \times 58.44 \text{ g/mol} = \mathbf{5.844} \text{ grams} \tag{Eq. 6.4}$$

On the SC-60, you type the formula **NaCl** into the tokenizer to retrieve its molecular weight (58.44), multiply by volume and concentration on the stack, and read the required mass in grams instantly.

6.2.2 Concept 2: The Conservation of Molecules (Serial Dilution: $C_1V_1 = C_2V_2$)

In laboratories, chemicals are rarely prepared from raw powders every day. Instead, labs keep concentrated **stock solutions** on the shelf (e.g. 5.0 M NaCl or 10× PBS buffer).

When you take a small pipette volume of stock solution (V_1) and add pure water to reach a larger final volume (V_2), the *concentration* drops, but the **total number of solute molecules remains strictly unchanged**:

$$\text{Moles of Solute} = C_1 \cdot V_1 = C_2 \cdot V_2 \tag{Eq. 6.5}$$

$$\text{Required Pipette Volume } V_1 = \frac{C_2 \cdot V_2}{C_1} \tag{Eq. 6.6}$$

The SC-60 features a dedicated [C₁V₁] softkey that solves for any of the four dilution variables in one step.

6.2.3 Concept 3: The Chemical Shock Absorber (pH & Buffers)

In 1909, Danish biochemist S.P.L. Sørensen was studying why beer fermentation fluctuated wildly between batches at the Carlsberg brewery. He discovered that enzymes function only within extremely narrow ranges of free hydrogen ion concentration $[H^+]$. Because $[H^+]$ can range from 1 mol/L down to 0.00000000000001 mol/L, Sørensen invented the negative logarithmic **pH scale**:

$$pH = -\log_{10}[H^+] \Leftrightarrow [H^+] = 10^{-pH} \quad \textbf{(Eq. 6.7)}$$

Every unit change on the pH scale represents a **tenfold multiplication** in acidity.

Why Pure Water Fails as a Buffer

Pure water at $pH = 7.0$ has virtually zero buffering capacity. If you add a single drop of strong hydrochloric acid to a liter of unbuffered water, the pH plunges from 7.0 down to 4.0, instantly destroying delicate biological proteins.

To maintain a stable pH, chemists use a **Buffer**: a mixture of a weak acid (HA) and its conjugate base salt (A^-). When acid is added, the conjugate base absorbs the protons; when base is added, the weak acid donates protons.

The equilibrium pH of a buffer is governed by the famous **Henderson-Hasselbalch equation**:

$$pH = pK_a + \log_{10}\left(\frac{[A^-]}{[HA]}\right) \quad \textbf{(Eq. 6.8)}$$

where $pK_a = -\log_{10}(K_a)$ is the acid dissociation constant (the pH at which the acid is exactly half-dissociated).

The SC-60's [BUFF] workspace automatically calculates the exact mass ratio of weak acid to conjugate base needed to hit any target pH.

6.2.4 Concept 4: Counting Molecules with Light (Beer-Lambert Law)

How do you measure the concentration of a clear, colorless protein or drug dissolved in water without evaporating the liquid?

You shine a beam of monochromatic light through the vial in a spectrophotometer and measure how much light emerges on the other side.

The **Beer-Lambert Law** states that the fraction of light absorbed (**Absorbance, *A***) is directly proportional to the concentration (c) of the absorbing molecules and the path length (b, standard 1.0 cm quartz cuvette):

$$A = -\log_{10}\left(\frac{\%T}{100}\right) = \varepsilon \cdot b \cdot c \quad \textbf{(Eq. 6.9)}$$

where ε (epsilon) is the molar absorptivity coefficient of the substance at that specific wavelength. By measuring absorbance on the SC-60, you calculate unknown solution concentrations instantly.

6.3 Real-World Field Walkthroughs

6.3.1 Field Case 1: Formulating a Phosphate Buffer (r/Chempros)

A biochemist needs to prepare 1.0 Liter of a 0.10 M Sodium Phosphate Buffer at physiological $pH = 7.40$.

- Conjugate Acid: Monobasic Sodium Phosphate (NaH_2PO_4, $MW = 119.98$ g/mol, $pK_a = 7.20$).
- Conjugate Base: Dibasic Sodium Phosphate (Na_2HPO_4, $MW = 141.96$ g/mol).

How many grams of each dry salt must be weighed on the balance?

How to Solve It on the SC-60

1. **Find the Base-to-Acid Ratio via Henderson-Hasselbalch:**

$$pH - pK_a = 7.40 - 7.20 = 0.20 \quad \textbf{(Eq. 6.10)}$$

$$\frac{[\text{Base}]}{[\text{Acid}]} = 10^{0.20} \approx \mathbf{1.585} \quad \textbf{(Eq. 6.11)}$$

2. **Determine Molar Proportions:**
   - Total concentration = [Acid] + [Base] = 0.10 M.
   - [Acid] × (1 + 1.585) = 0.10 ⇒ [Acid] = 0.10/2.585 = **0.0387 M**.
   - [Base] = 0.10 - 0.0387 = **0.0613 M**.
3. **Calculate Masses to Weigh for 1.0 Liter:**
   - Acid (NaH_2PO_4): 0.0387 mol × 119.98 g/mol = **4.64 grams**.
   - Base (Na_2HPO_4): 0.0613 mol × 141.96 g/mol = **8.70 grams**.

Weigh out 4.64 g of monobasic and 8.70 g of dibasic salt, dissolve in 800 mL pure water, and dilute to 1.0 L to achieve exactly pH 7.40!

6.3.2 Field Case 2: Serial Dilution from a 5 M Stock (r/chemistry)

A biology lab has a stock bottle of 5.0 M NaCl. A protocol requires 250 mL of a physiological saline solution (0.154 M NaCl). What volume of stock solution should be pipetted into a 250 mL volumetric flask?

Step-by-Step Mechanics

1. Use $C_1\ V_1 = C_2\ V_2$:

$$V_1 = \frac{C_2 \cdot V_2}{C_1} = \frac{0.154\ \text{M} \times 250\ \text{mL}}{5.0\ \text{M}} \quad \textbf{(Eq. 6.12)}$$

2. On the SC-60 stack:
   - **0.154** [ENTER] **250** [×] **5.0** [÷]
   - Register X displays: **7.70 mL**.

3. Pipette 7.70 mL of the 5 M stock solution into the flask and add distilled water up to the 250 mL mark.

6.4 Try It Yourself: Practice Problems

Problem 1: Molecular Weight of Glucose

Glucose has the chemical formula $C_6H_{12}O_6$. Using standard IUPAC atomic weights ($C = 12.011$, $H = 1.008$, $O = 15.999$), calculate the molecular weight of glucose. *(Solution: 6(12.011) + 12(1.008) + 6(15.999) = 72.066 + 12.096 + 95.994 =* ***180.156 g/mol****.)*

Problem 2: Beer-Lambert Concentration Check

A biochemist measures the light absorbance of a protein solution in a 1.0-cm cuvette at 280 nm. The spectrophotometer reads an absorbance $A = 0.650$. The protein's extinction coefficient is $\varepsilon = 45{,}000\ M^{-1}cm^{-1}$. Calculate the molar concentration $c = A / (\varepsilon \cdot b)$. *(Solution: c = 0.650 / (45000 × 1.0) =* ***1.444 × 10^{-5} M = 14.44 μM****.)*

Problem 3: Ideal Gas Volume at Standard Temperature

A reaction generates 0.25 moles of carbon dioxide gas at 25°C (298.15 K) and 1.0 atmosphere of pressure. Using the Ideal Gas Constant $R = 0.08206$ L·atm/(mol·K), calculate the gas volume in liters: $V = nRT / P$. *(Solution: V = (0.25 × 0.08206 × 298.15) / 1.0 =* ***6.115 Liters****.)*

SC-78 Commercial Bakery, Fermentation Science & Brewing Handbook

Who this is for: Anyone who loves cooking, artisan bread baking, or homebrewing, and wants to understand why professional bakeries treat flour as an immutable 100% anchor, and how dough water temperature is calculated like an engineering heat-transfer problem.

7.1 When the Kitchen Becomes a Laboratory

In home cooking, a recipe often reads: *"Add two cups of flour, one cup of milk, and a pinch of salt."*

If you walk into a commercial artisan bakery or craft brewery, you will never see a measuring cup. Volumetric measurements in baking are notoriously unreliable: depending on ambient humidity, how finely wheat was milled, and whether the flour was sifted or packed down, a single measuring cup of flour can weigh anywhere from **110 grams to 160 grams**—a massive 45% variance!

In bread making, where the difference between a dense sandwich loaf and a light, open-crumb rustic sourdough is a matter of a few percent water content, volume is the enemy of consistency. Everything must be weighed on a scale in **grams**.

Furthermore, professional kitchens face a constant mathematical puzzle: **scaling**. If your recipe makes three loaves of bread, but tomorrow morning you need to bake four hundred and fifty loaves, how do you scale the ingredients without altering the dough's hydration or salt balance?

To solve this, professional bakers invented a brilliant mathematical coordinate system called **Baker's Percentages (*BK*%)**. Combined with the thermodynamics of mixer motor friction and yeast fermentation kinetics, baking is not guesswork—it is delicious applied mathematics.

The **StackCalc 78 (SC-78)** brings this commercial precision to the kitchen bench, charcuterie curing cellar, and brew kettle. It incorporates locked Baker's Percentage scaling (**@BK%**), Desired Dough Temperature solvers (**TW**), equilibrium brining formulas, and brewing specific gravity calculations into a unified RPN stack.

7.1.1 Why Not Just Use a Smartphone?

- **The Flour Dust and Sticky Dough Reality:** A commercial bakery at 4:00 AM is coated in fine airborne flour dust and sticky dough residues. Trying to unlock a touchscreen phone with flour-coated thumbs or wet sourdough hands leads to missed touches, scratched glass, and dust-clogged ports. The SC-78 features tactile mechanical switches that can be operated through flour-covered palms and wiped down at the end of the shift.
- **Why Standard Math Percentages Fail:** In standard math, percentages must always sum to 100%. If you increase the water in a batch of bread from 65% to 75%, standard percentages would force the flour percentage to drop, which alters the calculated percentages of salt and yeast! In Baker's Percentages, **flour is permanently anchored at 100.0%**, keeping all ingredient relationships rock-solid.

7.2 Core Concepts: How Bakers and Brewers Think

7.2.1 Concept 1: Flour is the Anchor (Baker's Percentages)

In Baker's Percentages, the total weight of the flour is always defined as **100.0%**. If a formula uses multiple flours (e.g. 80% bread flour and 20% whole rye), their combined mass is 100.0%.

Every other ingredient is weighed as a direct ratio of the flour mass:

$$\text{Ingredient \%} = \left(\frac{\text{Mass of Ingredient}}{\text{Mass of Flour}}\right) \times 100 \quad \textbf{(Eq. 7.1)}$$

Classic Artisan Country Loaf Formula:

- **Flour Blend:** 1,000 g ⇒ **100.0%**
- **Water (Hydration):** 750 g ⇒ **75.0%**
- **Sourdough Starter (Levain):** 200 g ⇒ **20.0%**
- **Salt:** 20 g ⇒ **2.0%**
- **Total Formula Percent:** 100 + 75 + 20 + 2 = **197.0%**

Notice that the total percentage is 197%, not 100%! This system makes scaling effortless:

Scaling Down from Available Flour

If you open a bin and find you have exactly 42.5 kg of flour left, how much water and salt do you add?

- Water: 42.5 kg × 0.75 = **31.88 kg**
- Salt: 42.5 kg × 0.02 = **0.85 kg** = **850 grams**

Scaling Up from Target Finished Dough Weight

If you need to produce exactly 150 kg of finished dough to fill today's baguette orders:

$$\text{Required Flour Mass} = \frac{\text{Target Dough Mass}}{\text{Total Formula \%}/100} = \frac{150\text{ kg}}{1.97} = \mathbf{76.14}\text{ kg of Flour} \quad \textbf{(Eq. 7.2)}$$

Once you know the flour mass, every other ingredient scales in one keystroke on the SC-78.

7.2.2 Concept 2: The Heat of the Machine (Desired Dough Temperature)

Yeast is a living biological organism. Within normal kitchen temperatures, its metabolic fermentation rate **doubles for every 15°F (8.3°C) rise in temperature**.

For consistent flavor and structure, artisan bread dough should exit the mixer at an ideal target temperature—typically **76°F to 78°F (24°C to 26°C)**.

However, a commercial spiral mixer does not merely fold dough; its heavy rotating steel hook imparts intense **mechanical friction heat (15°F to 25°F)** directly into the dough mass.

To achieve the target dough temperature, the baker can only control one variable: **the temperature of the water (T_W) poured into the mixer bowl**.

The Three-Factor Rule

The baker measures:

- Room Temperature (T_{room})
- Flour Temperature (T_{flour}, measured with a probe thermometer)
- Mixer Friction Factor (F, determined experimentally for each mixer machine)

$$\text{Required Water Temperature } (T_W) = 3 \cdot T_{\text{desired}} - (T_{\text{room}} + T_{\text{flour}} + F) \quad \textbf{(Eq. 7.3)}$$

If the required water temperature calculates below 32°F, tap water cannot get cold enough, and the baker must substitute a portion of the water with crushed ice!

7.2.3 Concept 3: Density, Sugar, and Alcohol (Brewing Specific Gravity)

In brewing, beer is made by extracting fermentable sugars (maltose) from malted barley into hot water to create sweet liquid **wort**.

Brewers track fermentation using **Specific Gravity (*SG*)**: the density of the wort compared to pure water (1.000 at 20°C):

- Sweet unfermented wort typically has an **Original Gravity (*OG*)** between 1.040 and 1.080.
- As yeast consumes sugars, it converts dense sugar into less dense ethanol (density ≈ 0.789 g/mL) and carbon dioxide gas (which bubbles away).
- When fermentation ceases, the **Final Gravity (*FG*)** drops down to 1.008 to 1.015.

The percentage of **Alcohol by Volume (*ABV*)** is calculated from the gravity drop:

$$ABV\% \approx (OG - FG) \times 131.25 \quad \textbf{(Eq. 7.4)}$$

For example, a beer that starts at $OG = 1.056$ and finishes at $FG = 1.012$:

$$ABV = (1.056 - 1.012) \times 131.25 = 0.044 \times 131.25 = \mathbf{5.78\%}\ \text{ABV} \quad \textbf{(Eq. 7.5)}$$

7.3 Real-World Field Walkthroughs

7.3.1 Field Case 1: Summer Sourdough Water Temp & Ice Replacement (r/Sourdough)

A bakery on a warm July morning wants a finished dough temperature $T_D = 78$°F.

- Room Temperature: $T_R = 76$°F
- Flour Temperature: $T_F = 74$°F
- Spiral Mixer Friction Factor: $F = 22$°F
- Recipe calls for 15.0 kg total water.

1. What is the required water temperature (T_W)?
2. If tap water is 68°F, how many kilograms of ice must replace liquid water to reach the target temperature?

How to Solve It on the SC-78

1. **Calculate Required Water Temperature:**

$$T_W = 3(78) - (76 + 74 + 22) = 234 - 172 = \mathbf{62.0}^\circ\text{F} \quad \textbf{(Eq. 7.6)}$$

2. **Ice Mass Substitution Formula:** Because melting ice absorbs 144 BTU/lb (the latent heat of fusion):

$$\text{Ice Fraction} = \frac{T_{\text{tap}} - T_W}{T_{\text{tap}} + 112} = \frac{68 - 62}{68 + 112} = \frac{6}{180} = 0.0333 \Rightarrow \mathbf{3.33\%} \quad \textbf{(Eq. 7.7)}$$

$$\text{Ice Mass} = 15.0\text{ kg} \times 0.0333 = \mathbf{0.50}\text{ kg} = \mathbf{500}\text{ grams of ice} \quad \textbf{(Eq. 7.8)}$$

The baker weighs out 500 g of ice and 14.5 kg of tap water. When the ice melts during mixing, the dough exits the machine at exactly 78°F!

7.3.2 Field Case 2: Equilibrium Brining for Smoked Pork Belly (The Fresh Loaf)

A charcutier is curing a 3.5-kg slab of pork belly in a vacuum bag with 1.5-kg of water. The recipe calls for a balanced equilibrium salinity of 2.25% salt. How many grams of curing salt are required?

Step-by-Step Mechanics

1. In equilibrium brining, the salt dissolves and equilibrates evenly across the combined mass of meat and water:

$$\text{Total Mass} = 3.5\text{ kg (meat)} + 1.5\text{ kg (water)} = 5.0\text{ kg} \quad \textbf{(Eq. 7.9)}$$

2. Calculate salt mass at 2.25%:

   - **5000** [ENTER] **2.25** [%]
   - Register *X* displays: **112.5 grams of salt**.

The salt distributes completely through the meat with zero risk of over-salting!

7.4 Try It Yourself: Practice Problems

Problem 1: Baker's Percentage Sourdough Scaling

A recipe has the following Baker's Percentages:

- Flour: 100%
- Water: 72%
- Starter: 20%
- Salt: 2%

(Total Formula = 194%). You have an open bag of flour weighing exactly 35.0 kg. Calculate the required water mass and total expected dough yield. *(Solution: Water = 35.0 × 0.72 = **25.2 kg**. Total dough = 35.0 × 1.94 = **67.9 kg**.)*

Problem 2: Beer Alcohol by Volume (ABV)

A homebrewer measures a batch of Belgian Tripel with a hydrometer:

- Original Gravity: $OG = 1.082$
- Final Gravity: $FG = 1.014$

Calculate the Alcohol by Volume ($ABV\% = (OG - FG) \times 131.25$). *(Solution: 1.082 - 1.014 = 0.068. 0.068 × 131.25 =* ***8.925% ABV****.)*

Problem 3: Pan Volume Scaling

You have a cake recipe sized for an 8-inch round cake pan ($Area = \pi \cdot 4^2 = 50.27$ sq in). You want to bake the same cake in a 9 × 13-inch rectangular pan ($Area = 117.0$ sq in) at the same depth. Calculate the scaling multiplier: Multiplier = 117.0 / 50.27. *(Solution: 117.0 / 50.27 =* ***2.327×****. Multiply every ingredient in the recipe by 2.33!)*

SC-80 Clinical Nursing, Hospital Pharmacy & Infusions Handbook

Who this is for: Anyone who wants to understand the high-stakes applied mathematics of healthcare: how nurses, pharmacists, and paramedics turn complex physiological parameters into exact medication dosages and life-saving intravenous drip rates.

8.1 When a Decimal Point Means Everything

If you go to a pharmacy for a headache, the instructions on the bottle are simple: *"Take two tablets every six hours."*

In everyday over-the-counter medicine, drugs have a wide **therapeutic window**: the gap between the dose that cures your symptoms and the dose that makes you sick is comfortably broad.

Now step into an Intensive Care Unit (ICU) or a hospital oncology ward.

Here, medications are potent, narrow-therapeutic-index compounds: intravenous vasopressors that raise blood pressure by contracting blood vessels, cytotoxic chemotherapy agents that kill cancer cells, and powerful antibiotics like vancomycin. For these drugs, the therapeutic window is razor-thin:

- Give a patient **10% too little**, and a severe bloodstream infection overwhelms their immune system.
- Give a patient **10% too much**, and the drug permanently destroys their kidneys or induces life-threatening cardiac arrhythmias.

Furthermore, human bodies are not identical beakers. A 90-pound grandmother and a 250-pound athlete process drugs at vastly different metabolic speeds. Even more counterintuitive: two patients who have the exact same body weight can have completely different drug clearance rates if one is eighty years old and has declining kidney function.

In clinical practice, there is **zero safety margin for arithmetic error**. If a bridge engineer makes a 10% math error, structural steel has a safety factor of 3 to absorb the load. In neonatal intensive care, a single misplaced decimal point in a weight-based dopamine calculation (0.1 mg vs 0.01 mg) is a ten-fold overdose that can cause fatal cardiovascular collapse.

The **StackCalc 80 (SC-80)** is engineered specifically to eliminate clinical math errors. It introduces a persistent Patient Profile register (**@PAT**), Mosteller Body Surface Area (*BSA*) solvers, Cockcroft-Gault Creatinine Clearance (*CrCl*) calculators, and dedicated IV pump and gravity drip infusion keys directly into an immediate 4-level RPN stack.

8.1.1 Why Not Just Use a Smartphone?

- **The Code Blue Reality:** During an acute cardiac resuscitation (Code Blue) or rapid emergency septic shock treatment, alarms are sounding, arterial lines are flashing, and clinicians are wearing sterile surgical gloves, masks, and eye protection. Touchscreens often fail to register wet, gloved fingers, and face-unlock fails behind surgical masks. The SC-80 features crisp, tactile mechanical buttons with positive travel you can feel through nitrile gloves.

- **Infection Control & Cleanrooms:** Personal smartphones are notorious vectors of hospital-acquired bacterial infections (like MRSA). In sterile intravenous compounding cleanrooms (governed by USP 797 and USP 800 standards), taking a personal phone into a laminar flow hood is strictly prohibited. The SC-80 is an antimicrobial, non-communicating instrument that can be wiped down between shifts with hospital-grade disinfectant wipes.

8.2 Core Concepts: How Clinicians and Pharmacists Think

8.2.1 Concept 1: The Patient Profile (@PAT)

Clinical formulas repeatedly demand the same five core patient biometrics:

1. **Weight (W):** in kilograms (kg)
2. **Height (H):** in centimeters (cm)
3. **Age (A):** in years
4. **Serum Creatinine (S_{cr}):** in milligrams per deciliter (mg/dL)
5. **Biological Sex:** Male or Female

On a consumer calculator, a nurse or pharmacist must re-type these numbers into every single calculation, creating fatigue and endless opportunities for transposition typos.

On the SC-80, you enter the patient's biometrics once into the Row 0 softkeys ([W], [H], [AGE], [SCR], [SEX]). The calculator locks an active **Patient Profile (@PAT)** in memory. Every subsequent formula—whether Body Surface Area, Creatinine Clearance, or titration rates—automatically pulls the active profile without re-entry.

8.2.2 Concept 2: Why Surface Area Beats Weight (Mosteller BSA)

Why are cancer chemotherapy and burn resuscitation fluids dosed in **square meters of Body Surface Area (m^2)** rather than kilograms of weight?

Imagine two patients who both weigh 100 kg (220 lbs):

- Patient A is a 6-foot-4-inch muscular athlete.
- Patient B is a 5-foot-2-inch sedentary individual with significant adipose fat tissue.

Adipose fat tissue is relatively inactive metabolically: it has low blood flow and does not filter toxins. The athlete's liver and kidneys are much larger and receive substantially higher cardiac blood flow.

Physiological research shows that **cardiac output, renal glomerular filtration, and metabolic enzyme activity correlate directly with Body Surface Area (BSA)**, not raw fat weight.

In 1987, Dr. Richard Mosteller published a simple, elegant formula for estimating human body surface area from height and weight:

$$BSA = \sqrt{\frac{\text{Weight (kg)} \times \text{Height (cm)}}{3600}} \quad [\text{m}^2] \qquad \textbf{(Eq. 8.1)}$$

For an average adult (70 kg, 175 cm):

$$BSA = \sqrt{\frac{70 \times 175}{3600}} = \sqrt{\frac{12250}{3600}} = \sqrt{3.4028} \approx \mathbf{1.84}\ \text{m}^2 \quad \textbf{(Eq. 8.2)}$$

8.2.3 Concept 3: The Kidney Filter (Cockcroft-Gault Creatinine Clearance)

Most medications are cleared from the human bloodstream by the kidneys. If a patient's kidneys are failing, standard doses of drugs will accumulate to toxic, poisonous levels in the blood.

How do doctors measure kidney filtration speed without surgically probing the kidneys?

They measure **Creatinine**: a natural waste product produced by muscle breakdown. Healthy kidneys filter creatinine out of the blood into urine. If the kidneys are damaged, creatinine builds up in the bloodstream.

In 1976, Donald Cockcroft and Henry Gault developed the standard mathematical formula for estimating **Creatinine Clearance** ($\boldsymbol{CrCl}$):

$$\text{Male } CrCl = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{72 \times S_{cr}\ (\text{mg/dL})} \quad \left[\frac{\text{mL}}{\text{min}}\right] \quad \textbf{(Eq. 8.3)}$$

$$\text{Female } CrCl = \text{Male } CrCl \times \mathbf{0.85} \quad \textbf{(Eq. 8.4)}$$

Why Females Multiply by 0.85

On average, biological females have approximately 15% less muscle mass per kilogram of body weight than males, producing less baseline creatinine.

Why the Elderly Need Dose Reductions

Notice the numerator: (140 - Age). As humans age, functioning kidney nephrons naturally die off. An 85-year-old patient might have a "normal" looking blood creatinine reading of 1.0 mg/dL simply because their frail, shrinking muscle mass produces very little waste. But their actual filtration speed is less than half that of a young adult! The Cockcroft-Gault formula exposes this hidden renal decline so pharmacists can safely reduce antibiotic doses.

8.2.4 Concept 4: Counting Drops and Micrograms (IV Infusions)

In acute care, potent medications are delivered directly into the bloodstream through an intravenous catheter:

Volumetric Infusion Pumps (mL/hr)

A doctor orders a volume V (in mL) to infuse over time t (in hours):

$$\text{Pump Flow Rate} = \frac{V_{\text{mL}}}{t_{\text{hr}}} \quad \left[\frac{\text{mL}}{\text{hr}}\right] \quad \textbf{(Eq. 8.5)}$$

Gravity Drip Tubing (gtt/min)

In field ambulances or disaster areas without electronic pumps, nurses use gravity tubing calibrated to produce a specific number of drops per milliliter (the **Drop Factor**, DF, in gtt/mL):

$$\text{Drip Rate} = \frac{V_{\text{mL}} \times DF}{t_{\text{minutes}}} \quad \left[\frac{\text{drops}}{\text{min}}\right] \quad \textbf{(Eq. 8.6)}$$

Weight-Based Vasoactive Titrations (μg/kg/min)

For critical heart and blood pressure medications (e.g. norepinephrine or dopamine), the order is given per kilogram of patient weight per minute. To find the electronic pump rate:

$$\text{Pump Rate (mL/hr)} = \frac{\text{Ordered Dose } (\mu\text{g/kg/min}) \times \text{Weight (kg)} \times 60 \text{ min/hr}}{\text{IV Bag Concentration } (\mu\text{g/mL})} \quad \textbf{(Eq. 8.7)}$$

The SC-80's dedicated [TITR] softkey performs this three-step conversion in one touch.

8.3 Real-World Field Walkthroughs

8.3.1 Field Case 1: ICU Norepinephrine Shock Resuscitation (r/nursing)

A 75-kg patient in the Intensive Care Unit is in septic shock with a dangerously low Mean Arterial Pressure ($MAP < 55$ mmHg). The intensivist orders:

- **Norepinephrine (Levophed):** Titrate at **0.08 μg/kg/min**.
- **IV Bag Supply:** 4 mg (4,000 μg) reconstituted in a 250-mL bag of D_5W.

What volumetric rate in ***mL/hr*** should the nurse program into the IV infusion pump?

How to Solve It on the SC-80

1. **Find the IV Bag Drug Concentration:**
   - Concentration = 4000 μg / 250 mL = **16.0 μg/mL**.
2. **Compute Total Micrograms Per Minute:**
   - 0.08 μg/kg/min × 75 kg = **6.0 μg/minute**.
3. **Convert to Micrograms Per Hour:**
   - 6.0 μg/min × 60 min/hr = **360 μg/hour**.
4. **Compute Pump Rate (*mL/hr*):**
   - Rate = 360 μg/hr / 16.0 μg/mL = **22.5 mL/hr**.

On the SC-80 stack: **0.08** [ENTER] **75** [×] **60** [×] **16** [÷] → **22.5 mL/hr**.

8.3.2 Field Case 2: Geriatric Antibiotic Renal Dose Sizing (r/pharmacy)

A hospital clinical pharmacist is verifying an order for intravenous gentamicin in an 82-year-old female patient:

- Weight: 55.0 kg
- Serum Creatinine: 1.4 mg/dL

1. What is the patient's estimated Creatinine Clearance ($CrCl$)?

2. If the drug guidelines recommend reducing the dose when $CrCl < 50$ mL/min, does this patient require a dose reduction?

Step-by-Step Mechanics

1. **Apply the Cockcroft-Gault Equation for a Female:**

$$CrCl = \frac{(140 - 82) \times 55 \text{ kg}}{72 \times 1.4 \text{ mg/dL}} \times 0.85 = \frac{58 \times 55}{100.8} \times 0.85 = \frac{3190}{100.8} \times 0.85 = 31.65 \times 0.85 \approx \mathbf{26.9} \text{ mL/min} \quad \textbf{(Eq. 8.8)}$$

2. **Clinical Decision:** Because 26.9 mL/min is far below the 50 mL/min threshold, standard dosing would cause toxic accumulation. The pharmacist intervenes to extend the dosing interval from every 8 hours to every 24 hours, protecting the patient's kidneys!

8.4 Try It Yourself: Practice Problems

Problem 1: Mosteller Body Surface Area (BSA)

An oncology patient weighs 68.0 kg and has a height of 162.0 cm. Calculate the Body Surface Area: $BSA = \sqrt{((W \times H) / 3600)}$. *(Solution: 68 × 162 = 11,016. 11016 / 3600 = 3.060. √(3.060) =* ***1.749 m²****.)*

Problem 2: Gravity IV Drip Rate

A paramedic needs to infuse 1,000 mL of Normal Saline over 4 hours (240 minutes) during transport. The IV tubing has a drop factor of 15 gtt/mL. Calculate the gravity drip rate in drops per minute (*gtt*/min). *(Solution: Rate = (1000 × 15) / 240 = 15,000 / 240 =* ***62.5 ⇒ 62 to 63 drops/min****.)*

Problem 3: Mean Arterial Pressure (MAP)

A nurse measures an arterial line blood pressure of 110 / 70 mmHg ($SBP = 110$, $DBP = 70$). Calculate the Mean Arterial Pressure: $MAP = DBP + 1/3(SBP - DBP) = (2 \cdot DBP + SBP) / 3$. *(Solution: 2(70) + 110 = 140 + 110 = 250. 250 / 3 =* ***83.33 mmHg*** *(well above the critical 65 mmHg organ perfusion threshold!).)*

SC-95 Quantitative Finance, Actuarial Science & Six Sigma Handbook

Who this is for: Anyone who wants to understand the mathematics of capital, debt, and quality: how financial analysts value real estate and corporate acquisitions in their heads, and how industrial engineers use statistics to prevent factory defects.

9.1 The Cost of Tomorrow: Time, Uncertainty, and Capital

Here is a question with a seemingly obvious answer:

Everyone instinctively chooses the money today. But why?

Even if the future promise is 100% guaranteed by a government, money today is fundamentally more valuable than money in the future for two mathematical reasons:

1. **Opportunity Cost:** If you have the $10,000 today, you can invest it in a business or bank account earning compound interest. Over ten years at 7% annual return, that $10,000 doubles into almost $20,000.
2. **Inflation:** Over a decade, the rising price of goods erodes purchasing power; ten years from now, $10,000 buys substantially fewer groceries or gallons of fuel.

The mathematics of finance begins with this single realization: **cash flows occurring at different points in time cannot be added together.** A dollar in Year 10 is not the same currency as a dollar in Year 0. To compare them, you must translate future dollars backward into present dollars through **discounting**.

In corporate boardrooms, real estate acquisitions, and industrial factories, financial and quality decisions are made under severe time pressure:

- What monthly payment amortizes a $450,000 commercial mortgage at 6.5% interest?
- If a solar panel array costs $80,000 to install and saves $12,000 a year for a decade, what is its true Internal Rate of Return (IRR)?
- On an automotive stamping line producing ten thousand engine valves an hour, is the machinery capable of meeting tolerance limits without generating costly defective scrap?

In 1981, Hewlett-Packard introduced the **HP-12C**, designed with guidance from Turing Award winner Prof. William Kahan. It became the most enduring financial calculator in history, an icon carried into trading pits and boardrooms for decades.

The **StackCalc 95 (SC-95)** preserves this gold-standard financial workflow while fusing it with modern Six Sigma process capability indices (C_p, C_pk) and the complete continuous probability distribution suite of the statistical language **R** (**pnorm**, **qnorm**, **pt**, **qt**) directly within a 4-level RPN stack.

9.1.1 Why Not Just Use an Excel Spreadsheet?

- **The Negotiation and Boardroom Reality:** In a high-stakes commercial lending negotiation or corporate acquisition review, pulling up a laptop and fumbling through a messy spreadsheet with thirty nested tabs and broken circular references projects hesitation. A CFO or analyst who can verify an internal rate of return, bond yield to maturity, or balloon payment in six tactile clicks commands the room with authoritative confidence.
- **Strict Proctored Professional Exams:** The Chartered Financial Analyst (CFA) Institute, Certified Financial Planner (CFP) Board, and Society of Actuaries (SOA) enforce the strictest testing security in the academic world: laptops, tablets, smartphones, and graphing tools are completely prohibited from exam halls. Candidates must master a dedicated, non-communicating financial calculator.
- **Factory Floor Mobility:** An ASQ Six Sigma Black Belt auditing a vibrating manufacturing stamping floor cannot carry a laptop through oily walkways. One touch on [CPK] on the SC-95 immediately reports process capability and centering offsets right at the machine.

9.2 Core Concepts: How Financial Analysts and Quality Engineers Think

9.2.1 Concept 1: The 5-Key Time Value of Money (TVM) Engine

The foundation of all mortgages, car loans, leases, retirement accounts, and bond valuations rests on a single mathematical relationship connecting five fundamental variables:

$$PV \cdot (1+i)^N + PMT \cdot \left[\frac{(1+i)^N - 1}{i}\right] \cdot (1 + i \cdot S) + FV = 0 \qquad \textbf{(Eq. 9.1)}$$

- ***N*:** Total number of payment periods (e.g. 360 months for a 30-year mortgage).
- ***i*:** Interest rate per compounding period (e.g. 6.0% annual rate compounded monthly ⇒ $i = 6.0 \ / \ 12 = 0.50\%$).
- ***PV*:** Present Value (the lump-sum cash amount today).
- ***PMT*:** Periodic Payment (the recurring amount paid each period).
- ***FV*:** Future Value (the final balloon balance at the end).
- ***S*:** Timing switch (0 for **END** of period, 1 for **BEGIN** of period).

This equation is a five-way balance. If you know any four of the variables, the SC-95's TVM engine solves for the fifth unknown variable using Brent's root-finding algorithm.

The Cash-Flow Sign Convention (Money In vs. Money Out)

Why does William Kahan's TVM system require one variable to be negative?

Because **money cannot be created out of thin air**:

- **Cash received by you is POSITIVE (+).**
- **Cash paid out by you is NEGATIVE (-).**

If a bank lends you $300,000 to buy a home, that cash is paid to you today: $\boldsymbol{PV} = \boldsymbol{+300{,}000}$. Every month, when you write a mortgage check that leaves your bank account, that payment is an outflow: $\boldsymbol{PMT} = \boldsymbol{-1{,}896.20}$.

If you treat both as positive numbers, compound interest formulas collapse into nonsensical negative interest rates! Enforcing the sign convention reflects the true physics of accounting.

9.2.2 Concept 2: Discounted Cash Flows (NPV & IRR)

In corporate finance, companies invest money today to earn unpredictable streams of cash in the future.

Net Present Value (NPV)

To find out if a project is worth doing, you discount every future cash flow (CF_t) back to the present day using the company's required cost of capital rate (r):

$$NPV = CF_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \ldots + \frac{CF_k}{(1+r)^k} \quad \textbf{(Eq. 9.2)}$$

- If $\boldsymbol{NPV > 0}$, the investment generates more return than the cost of capital: **Do the project!**
- If $\boldsymbol{NPV < 0}$, the project destroys shareholder value: **Reject it!**

Internal Rate of Return (IRR)

The **Internal Rate of Return ($\boldsymbol{IRR}$)** is the exact annual interest rate at which NPV equals zero. It tells you the exact break-even yield of the project.

9.2.3 Concept 3: Taming the Bell Curve (Six Sigma C_{pk})

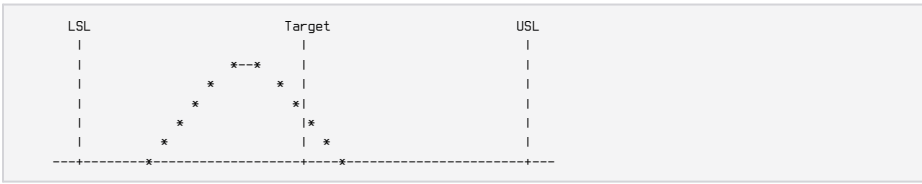
In industrial manufacturing, parts coming off an assembly line are never 100% identical. Natural vibrations, tool wear, and temperature fluctuations cause dimensions to vary in a Gaussian "bell curve" distribution with an average (μ) and standard deviation (σ).

Engineering blueprints define two boundary lines:

- **Lower Specification Limit ($\boldsymbol{LSL}$)**
- **Upper Specification Limit ($\boldsymbol{USL}$)**

Any part whose dimension falls outside these limits is defective scrap.

How do engineers evaluate whether a machine is capable of consistently meeting tolerances?



They use two **Process Capability Indices**:

Process Potential (C_p)

Measures the pure width of the process spread (6σ) relative to the allowable tolerance band:

$$C_p = \frac{USL - LSL}{6\sigma} \quad \textbf{(Eq. 9.3)}$$

If $C_p = 1.0$, the process width exactly equals the tolerance band. If $C_p > 1.33$, the process is capable of four-sigma or higher quality.

Process Capability Index (C_{pk})

What if the process is narrow, but its center has drifted toward one of the edges? C_p cannot detect this drift, but C_{pk} measures the distance from the average to the nearest specification limit:

$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right) \quad \textbf{(Eq. 9.4)}$$

In modern manufacturing:

- $C_{pk} < 1.0$**:** The process is actively producing defects.
- $C_{pk} = 1.33$**:** Industrial standard (4σ, fewer than 63 defects per million parts).
- $C_{pk} \geq 2.0$**:** World-Class Six Sigma (6σ, fewer than 3.4 defects per million parts!).

9.3 Real-World Field Walkthroughs

9.3.1 Field Case 1: 30-Year Mortgage Payment & Extra Principal (Bogleheads)

A homebuyer borrows $400,000 on a 30-year fixed mortgage at an annual interest rate of 6.50%.

1. What is the standard monthly principal and interest payment (PMT)?
2. If the homeowner pays an extra $300 per month toward principal ($PMT = \$2{,}828.31$), how many years earlier will the mortgage be paid off?

How to Solve It on the SC-95

1. **Calculate Standard Monthly Payment:**
   - Set mode to **END** of period.
   - Number of periods: $30 \times 12 = 360$ ⇒ **360** [↖] [N].
   - Monthly interest rate: $6.50\% \,/\, 12 = 0.54167\%$ ⇒ **6.5** [ENTER] **12** [÷] [↖] [I%YR].
   - Loan amount: **400000** [↖] [PV].
   - Final balance: **0** [↖] [FV].
   - Solve for payment: Tap [↖] [PMT].
   - Register X displays: **-$2,528.31** (Monthly payment).

2. **Calculate Accelerated Payoff with +$300 Extra:**

   - Enter new payment: **-2828.31** [↶] [PMT].
   - Solve for remaining periods: Tap [↶] [N].
   - Register *X* displays: **275.8 months** (23.0 years).
   - **Result:** Paying an extra $300/month cuts **7 full years (84 payments)** off the mortgage and saves over **$128,000 in interest**!

9.3.2 Field Case 2: Machining Tolerance Audit (C_{pk}) on a Stamping Line

A quality engineer audits a CNC shaft turning operation:

- Blueprint specification: 25.000 mm ± 0.050 mm ($LSL = 24.950$, $USL = 25.050$).
- A 50-part sample gives an average diameter $\mu = 25.012$ mm and standard deviation $\sigma = 0.008$ mm.

What is the process capability C_{pk}?

Step-by-Step Mechanics

1. **Compute Distance to Upper Limit:**

$$\frac{USL-\mu}{3\sigma} = \frac{25.050-25.012}{3(0.008)} = \frac{0.038}{0.024} \approx \mathbf{1.583} \qquad \textbf{(Eq. 9.5)}$$

2. **Compute Distance to Lower Limit:**

$$\frac{\mu-LSL}{3\sigma} = \frac{25.012-24.950}{3(0.008)} = \frac{0.062}{0.024} \approx \mathbf{2.583} \qquad \textbf{(Eq. 9.6)}$$

3. **Determine C_{pk}:**

$$C_{pk} = \min(1.583, 2.583) = \mathbf{1.58} \qquad \textbf{(Eq. 9.7)}$$

Because $C_{pk} = 1.58 > 1.33$, the process is capable and well-centered with virtually zero defect risk!

9.4 Try It Yourself: Practice Problems

Problem 1: Retirement Nest Egg (Future Value)

You invest $500 at the end of every month ($PMT = -500$) into an index fund averaging 8.0% annual return ($i = 8.0 / 12 = 0.6667\%$) for 35 years ($N = 35 \times 12 = 420$ months). You start with $0 ($PV = 0$). Calculate the accumulated Future Value (FV). *(Solution: FV = **$1,146,941.24**. A $500/month habit produces over $1.1 million dollars through compound interest!)*

Problem 2: Corporate Equipment Net Present Value (NPV)

A manufacturing plant is considering installing an automated packaging robot:

- Initial cost today: $CF_0 = -\$60{,}000$

- Cash savings Year 1: $CF_1 = +\$25{,}000$
- Cash savings Year 2: $CF_2 = +\$25{,}000$
- Cash savings Year 3: $CF_3 = +\$25{,}000$

The company's required discount rate is 10% ($r = 0.10$). Calculate the Net Present Value (NPV). *(Solution: Present value of annual savings = 25000 / 1.1 + 25000 / 1.1^2 + 25000 / 1.1^3 = 22727 + 20661 + 18783 = $62,171. NPV = -60000 + 62171 = +**$2,171.30**. Since NPV > 0, the project is financially sound and approved!)*

Problem 3: Standard Normal Distribution Quantile (Z-Score)

In statistical hypothesis testing, what is the two-tailed critical Z-score for a 95% confidence level (where 2.5% sits in the upper tail, meaning cumulative probability $P = 0.975$)? On the SC-95, enter **0.975** and tap [↗] [InvNorm] (**qnorm** in R). *(Solution: Register X displays **1.95996 ⇒ *1.96***. The iconic 1.96σ confidence interval multiplier!)*

Appendix A: Cross-Domain Governing Equations & Physical Constants

Who this is for: Practicing engineers, surveyors, tradespeople, lab analysts, clinicians, and quantitative analysts who need a rapid, single-glance desk reference of the governing mathematical equations, physical constants, and four-register stack (**T, Z, Y, X**) argument conventions across all nine instruments in the StackCalc Domain Series.

A.1 Universal RPN Stack & Precision Conventions

Across every instrument in the StackCalc Domain Series, all calculations execute inside a deterministic four-register operational stack (**T, Z, Y, X**) backed by IEEE 754 double-precision floating-point arithmetic (64-bit, 53-bit significand, ≈ 15–17 significant decimal digits):

Register	Operational Role	Behavior on Binary Operation (+, -, ×, ÷, y^x)	Behavior on ENTER
Register *T*	Top / 4th Level	Replicates downward into Z ($T' = T$) when stack drops	Receives prior Z ($T' = Z$)
Register *Z*	3rd Stack Level	Drops into Y ($Z = T$)	Receives prior Y ($Z = Y$)
Register *Y*	2nd Operand Level	Consumed as left operand ($Y \circ X$); receives prior Z	Replicates X ($Y' = X$)
Register *X*	Active Display Line	Consumed as right operand; receives result $f(Y, X)$	Remains in X (lift disabled for next digit)

A.2 Geomatics, Land Surveying & Flight Navigation (SC-35)

A.2.1 Sexagesimal Angle Conversion (D.MMSS)

Packed sexagesimal format encodes degrees (D), arc-minutes ($M \in [0, 59]$), and arc-seconds ($S \in [0, 59.99]$) as $D + M/100 + S/10000$. Conversion to decimal degrees (ϑ_{dec}) follows:

$$\theta_{\text{dec}} = D + \frac{M}{60} + \frac{S}{3600} \qquad \textbf{(Eq. A.1)}$$

A.2.2 Forward & Inverse Coordinate Geometry (COGO)

Given Station A at (N_1, E_1), horizontal distance D, and clockwise North azimuth α:

$$N_2 = N_1 + D\cos\alpha, \qquad E_2 = E_1 + D\sin\alpha \qquad \textbf{(Eq. A.2)}$$

Given two stations (N_1, E_1) and (N_2, E_2) with $\Delta N = N_2 - N_1$ and $\Delta E = E_2 - E_1$:

$$D = \sqrt{(\Delta N)^2 + (\Delta E)^2}, \qquad \alpha = \operatorname{atan2}(\Delta E, \Delta N) \pmod{360^\circ} \qquad \textbf{(Eq. A.3)}$$

A.2.3 Bowditch Compass Rule Traverse Adjustment

For a closed polygon traverse of perimeter $P = \sum D_i$ with linear misclosure $E_{mis} = \sqrt{((\sum \Delta N)^2 + (\sum \Delta E)^2)}$:

$$\delta N_i = -\Sigma\Delta N \cdot \frac{D_i}{P}, \qquad \delta E_i = -\Sigma\Delta E \cdot \frac{D_i}{P}, \qquad \text{Precision Ratio} = 1 : \frac{P}{E_{\text{mis}}} \qquad \textbf{(Eq. A.4)}$$

A.2.4 Aviation Wind Triangle, Crosswind & Density Altitude

With runway heading ψ, wind direction ϑ_w, wind speed V_w, and true airspeed V_{TAS}:

$$V_{\text{XW}} = V_w \sin(\theta_w - \psi), \qquad V_{\text{HW}} = V_w \cos(\theta_w - \psi), \qquad \text{WCA} = \arcsin\!\left(\frac{V_{\text{XW}}}{V_{\text{TAS}}}\right) \tag{Eq. A.5}$$

$$\text{PA} = \text{Elev} + 1000 \times (29.92 - \text{Alt}_{\text{inHg}}), \qquad \text{DA} \approx \text{PA} + 120 \times (T_{\text{OAT}} - T_{\text{ISA}}) \tag{Eq. A.6}$$

where standard temperature at altitude is $T_{\text{ISA}} = 15°\text{C} - 2°\text{C} \times (\text{PA} / 1000\ \text{ft})$.

A.3 Industrial Trades, Electrical & Mechanical (SC-14)

A.3.1 Framing Trigonometry & Hip/Valley Rafters

For a roof with unit rise U (inches per 12 in of horizontal run) and span S (where run $R = S/2$):

$$\theta = \arctan\!\left(\frac{U}{12}\right), \qquad L_{\text{common}} = R\sqrt{1 + \left(\frac{U}{12}\right)^2}, \qquad L_{\text{hip}} = R\sqrt{2 + \left(\frac{U}{12}\right)^2} \tag{Eq. A.7}$$

A.3.2 Electrical Conduit Offset Bends & Shrink

To clear an obstruction of height O using two equal bends of angle ϑ:

$$L_{\text{travel}} = O \cdot \csc\theta = \frac{O}{\sin\theta}, \qquad S_{\text{shrink}} = O \cdot \tan\!\left(\frac{\theta}{2}\right) \tag{Eq. A.8}$$

Bend Angle (ϑ)	Travel Multiplier (csc ϑ)	Shrink per Inch of Offset (tan(ϑ/2))
10°	5.759 × O	0.087 in/in (≈ 1/16 in)
22.5°	2.613 × O	0.199 in/in (≈ 3/16 in)
30°	2.000 × O	0.268 in/in (≈ 1/4 in)
45°	1.414 × O	0.414 in/in (≈ 3/8 in)
60°	1.155 × O	0.577 in/in (≈ 9/16 in)

A.3.3 Three-Phase AC Power & NEC Conductor Voltage Drop

For a balanced three-phase load with line-to-line voltage V_{LL}, line current I_L, power factor $\cos\varphi$, one-way run length L (feet), conductor circular mils CM, and resistivity constant K (12.9 Ω·CM/ft for copper, 21.2 Ω·CM/ft for aluminum):

$$P_{3\phi} = \sqrt{3} \cdot V_{LL} \cdot I_L \cdot \cos\phi, \qquad V_{\text{drop},3\phi} = \frac{\sqrt{3} \cdot K \cdot I_L \cdot L}{\text{CM}}, \qquad V_{\text{drop},1\phi} = \frac{2 \cdot K \cdot I_L \cdot L}{\text{CM}} \tag{Eq. A.9}$$

A.4 Systems Programming, Embedded & Cyber Operations (SC-16)

A.4.1 Two's Complement Signed Integer Representation

In an N-bit word ($N \in \{8, 16, 32, 64\}$), a bit vector $b_{N-1} b_{N-2} \ldots b_1 b_0$ represents the signed integer:

$$V_{\text{signed}} = -b_{N-1} 2^{N-1} + \sum_{k=0}^{N-2} b_k 2^k, \qquad -x \equiv (\sim x + 1) \pmod{2^N} \tag{Eq. A.10}$$

A.4.2 IEEE 754 Single-Precision (32-Bit) Floating-Point Layout

For sign bit $s \in (0, 1)$, 8-bit biased exponent $E \in [1, 254]$, and 23-bit fractional mantissa M:

$$x = (-1)^{s} \times \left(1 + \frac{M}{2^{23}}\right) \times 2^{E-127} \tag{Eq. A.11}$$

A.4.3 Cryptographic Modular Exponentiation, Shannon Entropy & CIDR

Fast square-and-multiply modular exponentiation (MOD, y^x):

$$C = M^{e} \pmod{n} \tag{Eq. A.12}$$

Shannon information entropy $H(X)$ (bits per symbol) and IPv4 CIDR usable host capacity for prefix $/P$:

$$H(X) = -\sum_{i=1}^{m} p_i \log_2 p_i, \qquad N_{\text{hosts}} = 2^{32-P} - 2 \tag{Eq. A.13}$$

A.5 Cinema Optics, Broadcast Timecode & Sound (SC-24)

A.5.1 SMPTE Timecode & NTSC 29.97 Drop-Frame Compensation

Actual NTSC broadcast frame rate is $f_{\text{NTSC}} = 30 \times 1000 / 1001 \approx 29.97002997$ fps. Over one hour, a nominal 30 fps counter drifts by $+108$ frames ($+3.6$ s). Drop-Frame (**29.97 DF**) drops frame numbers **:00** and **:01** at the start of every minute except minutes divisible by 10:

$$N_{\text{dropped/hr}} = 60 \times 2 - 6 \times 2 = 108 \text{ frames/hr} \tag{Eq. A.14}$$

A.5.2 Hyperfocal Distance, Depth of Field & Rotary Shutter Exposure

For lens focal length f, aperture N (f-stop or T-stop), circle of confusion c, and focus distance d:

$$H = \frac{f^2}{N \cdot c} + f, \qquad D_{\text{near}} = \frac{H \cdot d}{H + (d - f)}, \qquad D_{\text{far}} = \frac{H \cdot d}{H - (d - f)} \quad (d < H) \tag{Eq. A.15}$$

For frame rate FPS and rotary shutter angle $\vartheta_{\text{shutter}}$ (degrees):

$$t_{\text{exp}} = \frac{\theta_{\text{shutter}}}{360^{\circ} \times \text{FPS}} \tag{Eq. A.16}$$

A.6 RF Spectrum, Transmission Lines & Microwave (SC-50)

A.6.1 Decibel Power & Voltage Ratios

$$G_{\text{dB}} = 10 \log_{10}\left(\frac{P_2}{P_1}\right) = 20 \log_{10}\left(\frac{V_2}{V_1}\right), \qquad P_{\text{dBm}} = 10 \log_{10}\left(\frac{P_{\text{watts}}}{1\ \text{mW}}\right) \tag{Eq. A.17}$$

A.6.2 Reflection Coefficient (Γ), VSWR, Return Loss & Mismatch Loss

For characteristic impedance Z_0 and load impedance Z_L:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \qquad \mathrm{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \qquad \mathrm{RL}_{\mathrm{dB}} = -20\log_{10}|\Gamma|, \qquad \mathrm{ML}_{\mathrm{dB}} = -10\log_{10}(1 - |\Gamma|^2) \tag{Eq. A.18}$$

A.6.3 Friis Free-Space Path Loss & Thermal Noise Floor

$$\mathrm{FSPL}_{\mathrm{dB}} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\tfrac{4\pi}{c}\right), \qquad N_{\mathrm{dBm}} = -174\ \mathrm{dBm/Hz} + 10\log_{10}(B_{\mathrm{Hz}}) + \mathrm{NF}_{\mathrm{dB}} \tag{Eq. A.19}$$

A.7 Wet-Lab Chemistry, Stoichiometry & Spectroscopy (SC-60)

A.7.1 Molarity, Dilutions & Beer-Lambert Spectrophotometry

$$M = \frac{m_{\mathrm{g}}}{\mathrm{MW}_{\mathrm{g/mol}} \times V_{\mathrm{L}}}, \qquad C_1 V_1 = C_2 V_2, \qquad A = -\log_{10}\left(\frac{I}{I_0}\right) = \varepsilon \cdot \ell \cdot c \tag{Eq. A.20}$$

A.7.2 pH, Henderson-Hasselbalch Buffers & Arrhenius Kinetics

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+], \qquad \mathrm{pH} = \mathrm{p}K_a + \log_{10}\left(\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}\right), \qquad \ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \tag{Eq. A.21}$$

where the universal gas constant is $R = 8.3144626\ \mathrm{J{\cdot}mol^{-1}{\cdot}K^{-1}}$ and Avogadro's constant is $N_A = 6.02214076 \times 10^{23}\ \mathrm{mol^{-1}}$.

A.8 Commercial Bakery, Fermentation Science & Brewing (SC-78)

A.8.1 Baker's Percentage & Desired Dough Temperature (DDT)

$$\text{Baker's }\%_i = \frac{W_i}{W_{\text{total flour}}} \times 100\%, \qquad T_{\text{water}} = 3 \times \mathrm{DDT} - (T_{\text{room}} + T_{\text{flour}} + F_{\text{friction}}) \tag{Eq. A.22}$$

A.8.2 Brewing Gravity, Alcohol by Volume (ABV), Strike Water & IBUs

$$\mathrm{ABV}\% = (\mathrm{OG} - \mathrm{FG}) \times 131.25, \qquad T_{\text{strike}} = \frac{0.2}{R_{\text{qt/lb}}}(T_{\text{target}} - T_{\text{grain}}) + T_{\text{target}} \tag{Eq. A.23}$$

$$\mathrm{IBU} = \frac{W_{\text{oz}} \times \mathrm{AA}\% \times U_{\text{Tinseth}} \times 7489}{V_{\text{gal}}} \tag{Eq. A.24}$$

A.9 Clinical Nursing, Hospital Pharmacy & Infusions (SC-80)

A.9.1 Body Surface Area (Mosteller) & Ideal / Adjusted Body Weight

$$\mathrm{BSA}\ (\mathrm{m^2}) = \sqrt{\frac{\mathrm{Height_{cm}} \times \mathrm{Weight_{kg}}}{3600}}, \qquad \mathrm{ABW_{kg}} = \mathrm{IBW_{kg}} + 0.4(\mathrm{Actual_{kg}} - \mathrm{IBW_{kg}}) \tag{Eq. A.25}$$

where Devine Ideal Body Weight is $\mathrm{IBW_{male}} = 50 + 2.3(H_{\text{in}} - 60)$ and $\mathrm{IBW_{female}} = 45.5 + 2.3(H_{\text{in}} - 60)$.

A.9.2 Cockcroft-Gault Creatinine Clearance & Infusion Pump Rates

$$\text{CrCl (mL/min)} = \frac{(140 - \text{Age}_{\text{yr}}) \times \text{Weight}_{\text{kg}}}{72 \times \text{SCr}_{\text{mg/dL}}} \times (0.85 \text{ if female}) \tag{Eq. A.26}$$

$$\text{Rate (mL/hr)} = \frac{\text{Dose}_{\text{mcg/kg/min}} \times \text{Weight}_{\text{kg}} \times 60 \text{ min/hr}}{\text{Concentration}_{\text{mcg/mL}}}, \qquad \text{Drops/min} = \frac{V_{\text{mL}} \times D_{\text{gtt/mL}}}{t_{\text{min}}} \tag{Eq. A.27}$$

A.10 Quantitative Finance, Actuarial Science & Six Sigma (SC-95)

A.10.1 Compound Time Value of Money (TVM) & Net Present Value (NPV)

$$PV(1+i)^N + PMT \cdot \frac{(1+i)^N - 1}{i} \cdot (1 + i \cdot \text{mode}) + FV = 0, \qquad \text{NPV} = CF_0 + \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t} \tag{Eq. A.28}$$

A.10.2 Six Sigma Process Capability (C_{p}, C_{pk}) & Black-Scholes Option Pricing

$$C_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}}, \qquad C_{pk} = \min\left(\frac{\text{USL} - \mu}{3\hat{\sigma}}, \frac{\mu - \text{LSL}}{3\hat{\sigma}}\right) \tag{Eq. A.29}$$

$$C = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{Eq. A.30}$$

Appendix B: Master Row 0 Softkey & Shifted Keycap Directory

Who this is for: Operators, educators, and engineers who want a complete hardware keycap and Row 0 domain softkey directory across all nine StackCalc Domain Series instruments, showing exact pre-execution stack arguments (Z, Y, X) and post-execution outputs (Y', X').

B.1 Shared Core Keys Across All Nine Instruments

Every calculator in the StackCalc Domain Series shares the same physical core keypad layout for stack manipulation, register storage, and arithmetic:

Keycap / Sequence	Function Name	Pre-Execution Stack (Y, X)	Post-Execution Stack (Y', X')
ENTER	Stack Lift & Duplicate	**Y, X**	**X, X** (prior $Y \to Z$, $Z \to T$)
x ⇄ y	Swap Registers X and Y	**Y, X**	**X, Y** (Z and T unchanged)
R↓	Roll Stack Downward	**T, Z, Y, X**	**X, T, Z, Y**
⌫ / C ON	Backspace / Clear X	**Y, X**	**Y, 0.0000** (disables stack lift)
+/−	Change Sign / Negate	**Y, X**	**Y, -X**
STO 0–9	Store X in Register R_n	**Y, X**	**Y, X** ($R_n \leftarrow X$)
RCL 0–9	Recall Register R_n to X	**Y, X**	**X, R$_n$** (lifts stack first)
↰	Yellow Primary Shift	—	Arms yellow shifted legend for next key
↱	Blue Secondary Shift	—	Arms blue shifted legend for next key
SHOW	Full-Precision Mantissa Peek	**Y, X**	Temporarily reveals all 15 significant digits of X

B.2 SC-35 Geomatics, Land Surveying & Flight Navigation

Keycap / Softkey	Layer	Operation & Stack Signature
FWD	Row 0 Softkey (Col 0)	**Forward Traverse:** Consumes distance D (Y) and azimuth α (X); returns ΔE in Y' and ΔN in X'.
INV	Row 0 Softkey (Col 1)	**Coordinate Inverse:** Consumes ΔN (Y) and ΔE (X); returns azimuth $\alpha \in [0°, 360°)$ in Y' and horizontal distance D in X'.
BRG	Row 0 Softkey (Col 2)	**Quadrant Bearing:** Formats azimuth in X as surveyor quadrant bearing (**NE**, **SE**, **SW**, **NW**) and degree-minute-second components.
WND	Row 0 Softkey (Col 3)	**E6B Wind Triangle:** Decomposes wind speed/angle into headwind (Y) and crosswind (X) components, or solves Wind Correction Angle (**WCA**).
ISA	Row 0 Softkey (Col 4)	**ISA & Density Altitude:** Computes Pressure Altitude (**PA**) and Density Altitude (**DA**) from elevation, altimeter setting, and OAT.
trav	Row 0 Softkey (Col 5)	**Bowditch Traverse Accumulator:** Accumulates station leg (ΔN_i, ΔE_i, D_i) and computes closure error and precision ratio.
↰ ▸HR	Yellow Shift	Converts packed sexagesimal **D.MMSS** in X to decimal degrees/hours.
↱ ▸HMS	Blue Shift	Converts decimal degrees/hours in X to packed sexagesimal **D.MMSS**.

Keycap / Softkey	Layer	Operation & Stack Signature
[→] [POL]	Blue Shift	Rectangular (Y, X) to Polar: magnitude $r \to X$, angle $\vartheta \to Y$.
[←] [REC]	Yellow Shift	Polar (Y, X) to Rectangular: $x = r\cos\vartheta \to X$, $y = r\sin\vartheta \to Y$.

B.3 SC-14 Industrial Trades, Electrical & Mechanical

Keycap / Softkey	Layer	Operation & Stack Signature
[RUN]	Row 0 Softkey (Col 0)	Stores or solves horizontal framing run R in feet-inches-fractions (**FT-IN**).
[RISE]	Row 0 Softkey (Col 1)	Stores or solves vertical rise from unit pitch (U/12) and run R.
[DIAG]	Row 0 Softkey (Col 2)	Computes common rafter diagonal $\sqrt{(\text{Run}^2 + \text{Rise}^2)}$ in **FT-IN** ([→] [HIP] for hip/valley).
[BEND]	Row 0 Softkey (Col 3)	**Conduit Offset Solver:** Consumes offset O (Y) and bend angle ϑ (X); returns travel length L_{travel} in X and shrink S in Y.
[VD]	Row 0 Softkey (Col 4)	**NEC Voltage Drop:** Computes conductor voltage drop V_{drop} and terminal voltage from current I, one-way distance L, and AWG wire gauge.
[TBL]	Row 0 Softkey (Col 5)	**Trade Lookup Table:** Recalls NEC Chapter 9 circular mils / ampacity and ANSI drill-tap allowances.
[←] [3PH÷]	Yellow Shift	Divides apparent three-phase power S (Y) by $\sqrt{3} \cdot V_{LL}$ (X) to return line current I_L in X.
[←] [→mm]	Yellow Shift	Converts decimal or fractional inches in X to exact millimeters ($\times$ 25.4).
[→] [▶in]	Blue Shift	Converts millimeters in X to decimal and nearest 1/16 in fractional inches.

B.4 SC-16 Systems Programming, Embedded & Cyber Operations

Keycap / Softkey	Layer	Operation & Stack Signature
[HEX]	Row 0 Softkey (Col 0)	Switches display and keypad entry to base-16 hexadecimal (**0**–**9**, **A**–**F**).
[DEC]	Row 0 Softkey (Col 1)	Switches display to signed/unsigned base-10 decimal integer or float mode.
[BIN]	Row 0 Softkey (Col 2)	Switches display to base-2 binary bit-grouped view ([→] [OCT] for base-8 octal).
[WSZ]	Row 0 Softkey (Col 3)	Cycles hardware word width across **8-bit**, **16-bit**, **32-bit**, and **64-bit** registers.
[ENDN]	Row 0 Softkey (Col 4)	Swaps byte endianness (**BSWAP16**, **BSWAP32**, **BSWAP64**) of integer in X.
[FLT]	Row 0 Softkey (Col 5)	Reinterprets 32-bit hex word in X as an IEEE 754 **float32** value (and vice versa).
[←] [AND] / [OR] / [XOR]	Yellow Shift	Bitwise logical **AND**, **OR**, or **XOR** between Y and X masked to active word width.
[←] [NOT]	Yellow Shift	Bitwise one's complement ($\sim X$) masked to active word width ([+/−] for two's complement).
[→] [SL] / [SR] / [ASR]	Blue Shift	Logical shift left ($<<$), logical shift right ($>>>$), and arithmetic shift right ($>>$).

Keycap / Softkey	Layer	Operation & Stack Signature
[→] [RL] / [RR]	Blue Shift	Circular bit rotate left (**ROL**) and rotate right (**ROR**) by X bit positions.
[MOD]	Primary Key	Computes Y mod X ([MOD] followed by [y^x] executes Z^Y mod X).

B.5 SC-24 Cinema Optics, Broadcast Timecode & Sound

Keycap / Softkey	Layer	Operation & Stack Signature
[TC]	Row 0 Softkey (Col 0)	Toggles **HH:MM:SS:FF** SMPTE timecode entry and arithmetic mode.
[FPS]	Row 0 Softkey (Col 1)	Cycles frame rate across **23.976**, **24.00**, **25.00**, **29.97 DF**, **29.97 NDF**, **30.00**, and **60.00** fps.
[DOF]	Row 0 Softkey (Col 2)	**Depth of Field Solver:** Consumes focal length f (Z), T-stop (Y), and focus distance d (X); returns hyperfocal H, near limit D_{near}, and far limit D_{far}.
[EXP]	Row 0 Softkey (Col 3)	**Exposure & Shutter Solver:** Computes exposure time t_{exp} or T-stop compensation across FPS, shutter angle, and ND filters.
[AUD]	Row 0 Softkey (Col 4)	**Acoustic Delay & Wavelength:** Computes sound propagation delay (ms) and frame offset for distance d at ambient temperature T.
[dB⊕]	Row 0 Softkey (Col 5)	**Incoherent Decibel Power Sum:** Computes $10\log_{10}(10^{Y/10} + 10^{X/10})$ for combining two audio/noise sources in dB.

B.6 SC-50 RF Spectrum, Transmission Lines & Microwave

Keycap / Softkey	Layer	Operation & Stack Signature
[dBm]	Row 0 Softkey (Col 0)	Converts power in Watts/milliwatts in X to dBm ($10\log_{10}(P$ / 1 mW)).
[→W]	Row 0 Softkey (Col 1)	Converts power level in dBm in X to linear power in Watts ($10^{(X-30)/10}$).
[SWR]	Row 0 Softkey (Col 2)	Converts load impedance Z_L or reflection magnitude $\lvert\Gamma\rvert$ in X to standing wave ratio VSWR.
[FSPL]	Row 0 Softkey (Col 3)	**Friis Path Loss:** Consumes link distance d (Y) and frequency f (X); returns free-space path loss FSPL_{dB} in X.
[SMTH]	Row 0 Softkey (Col 4)	**L-Network Impedance Matcher:** Consumes Z_0 (Z), R_L (Y), and X_L (X); returns series/shunt matching reactances and L, C values.
[NF]	Row 0 Softkey (Col 5)	**Friis Cascade Noise Figure & Thermal Floor:** Computes cascaded noise figure or receiver noise floor N_{dBm} for bandwidth B.
[↶] [RL] / [ML]	Yellow Shift	Converts VSWR or $\lvert\Gamma\rvert$ in X directly to Return Loss (RL_{dB}) or Mismatch Loss (ML_{dB}).

B.7 SC-60 Wet-Lab Chemistry, Stoichiometry & Spectroscopy

Keycap / Softkey	Layer	Operation & Stack Signature
[MW]	Row 0 Softkey (Col 0)	Computes or recalls molecular weight (g/mol) from atomic masses ([C ON], [H], [N], [O], **[Na]**, **[Cl]**, [P], [S]).
[MOL]	Row 0 Softkey (Col 1)	Solves mass $m_g = M \times \text{MW} \times V_{\text{L}}$ or molarity M from mass, molecular weight, and volume.

Keycap / Softkey	Layer	Operation & Stack Signature
C₁V₁	Row 0 Softkey (Col 2)	**Dilution Solver:** Consumes stock C_1 (Z), target C_2 (Y), and target volume V_2 (X); returns stock aliquot V_1 in X' and diluent volume V_2 - V_1 in Y'.
pH	Row 0 Softkey (Col 3)	Converts $[H^+]$ molarity in X to pH = -log₁₀(X) ([↗] [10□ˣ] for inverse).
BUFF	Row 0 Softkey (Col 4)	**Henderson-Hasselbalch Buffer:** Consumes pK_a (Z), target pH (Y), and total molarity C_{tot} (X); returns [HA] and [A⁻].
ABS	Row 0 Softkey (Col 5)	**Beer-Lambert Law:** Converts percent transmittance $\%T$ to absorbance $A = 2$ - log₁₀($\%T$) or solves concentration $c = A$ / (ε ℓ).
TITR	Shifted Solver	Computes equivalence-point volume V_{eq} and titration curve pH from titrant and analyte volumes/molarities.

B.8 SC-78 Commercial Bakery, Fermentation Science & Brewing

Keycap / Softkey	Layer	Operation & Stack Signature
FLR	Row 0 Softkey (Col 0)	Sets total flour weight (100% Baker's Percentage reference base) from Register X.
BKR%	Row 0 Softkey (Col 1)	Converts ingredient percentage in X to batch weight (or batch weight to Baker's %).
DDT	Row 0 Softkey (Col 2)	**Desired Dough Temperature:** Consumes target DDT (T), room temp (Z), flour temp (Y), and mixer friction (X); returns required water temperature T_{water}.
ABV	Row 0 Softkey (Col 3)	**Fermentation Gravity Solver:** Consumes Original Gravity OG (Y) and Final Gravity FG (X); returns ABV% in X' and apparent attenuation in Y'.
STRK	Row 0 Softkey (Col 4)	**Mash Strike Water:** Consumes grist ratio R (Z), grain temp (Y), and target mash rest temp (X); returns strike water temperature T_{strike}.
IBU	Row 0 Softkey (Col 5)	**Tinseth Hop Bitterness:** Computes International Bitterness Units (IBU) from hop mass, alpha-acid %, boil time, wort gravity, and batch volume.

B.9 SC-80 Clinical Nursing, Hospital Pharmacy & Infusions

Keycap / Softkey	Layer	Operation & Stack Signature
BSA	Row 0 Softkey (Col 0)	**Mosteller BSA:** Consumes patient height in cm (Y) and weight in kg (X); returns Body Surface Area (m^2) in X'.
IBW	Row 0 Softkey (Col 1)	**Devine Ideal & Adjusted Body Weight:** Computes IBW_{kg} and ABW_{kg} from patient sex, height, and actual weight.
CrCl	Row 0 Softkey (Col 2)	**Cockcroft-Gault Renal Clearance:** Consumes age (Z), weight in kg (Y), and serum creatinine SCr (X); returns CrCl (mL/min).
MCG	Row 0 Softkey (Col 3)	**Weight-Based IV Titration:** Converts ordered dose (mcg/kg/min), patient weight (kg), and bag concentration (mcg/mL) to pump rate (mL/hr).
GTT	Row 0 Softkey (Col 4)	**Gravity IV Drip Rate:** Consumes infusion rate in mL/hr (Y) and tubing drop factor in gtt/mL (X); returns rounded whole drops per minute (gtt/min).
PED	Row 0 Softkey (Col 5)	**Pediatric Safe-Dose Guardrail:** Verifies ordered dose against mg/kg/day safe range and computes liquid suspension volume (mL/dose).

B.10 Quantitative Finance, Actuarial Science & Six Sigma (SC-95)

Keycap / Softkey	Layer	Operation & Stack Signature
[N] / [I/Y] / [PV] / [PMT] / [FV]	Row 0 Softkeys (Cols 0–4)	**Five-Key TVM Solver:** If preceded by numeric entry, stores X into that TVM variable; if pressed after another TVM key, solves for that variable using signed cash-flow convention.
[CPK]	Row 0 Softkey (Col 5)	**Six Sigma Process Capability:** Consumes LSL (T), USL (Z), sample mean μ (Y), and standard deviation $\hat{\sigma}$ (X); returns C_{pk} in X and C_p in Y.
[↰] [I%YR]	Yellow Shift	Divides annual nominal rate in Y by compounding periods per year in X and stores periodic rate in [I/Y].
[↰] [AMRT]	Yellow Shift	Computes principal paid (ΔPrin), interest paid (ΔInt), and remaining balance over payment range $[p_1, p_2]$.
[↰] [NPV] / [IRR]	Yellow Shift	Computes Net Present Value or Internal Rate of Return across cash-flow series CF_0, CF_1, ..., CF_T.
[↱] [NormCDF] / [InvNorm]	Blue Shift	Standard normal cumulative probability $\Phi(z)$ and inverse quantile $\Phi^{-1}(p)$.

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